N. Robertson and P. D. Seymour, Graph Minors. XVI. Excluding a Non-Planar Graph, *J. Combin. Theory Ser. B* **89** (2003), 43–76.

About this copy

This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM16.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Misprint.** Section 1, page 45: in the definition of an r-ring, $t_i \in X_1$ becomes $t_i \in X_i$.
2. **Misprint.** Section 2, page 47: in the definition of a design, "in" becomes "is".
3. **Missing proof.** Proof of 2.2, page 48: the proof is added.
4. **Statement mistake.** Section 2, page 48: the sentence before 2.3 is restricted to the case $Z \cap V(H) = \emptyset$, and a weaker form is added for the general case.
5. **Proof gap.** Statement of 2.3, page 48: the second case is weakened in the same way.
6. **Misprint.** Proof of 1.3, page 51: B becomes B_i^j in the definition of C_i^j.

7. **Missing proof.** Proof of 1.3, page 51: the proof that $\mathcal{L}$ is a location whose design belongs to $\mathcal{D}$ is added.
8. **Proof gap.** Proof of 1.3, page 51: a separation of order at most 2 is added to $\mathcal{L}^+$ for each edge with an end in Z and all its ends in Z or in the core of $\mathcal{L}$.
9. **Proof gap.** Proof of 1.3, page 52: the last step uses the weaker forms of page 48, and r-rings become $(w+1)r$-rings.
10. **Misprint.** Proof of 3.1, page 54: $i \leq j \leq n$ becomes $i < j \leq n$.
11. **Misprint.** Proof of 5.5, page 57: the letter l becomes the digit 1.
12. **Misprint.** Proof of 6.3, page 60: $X \cup \{x\}$ becomes $X \cup \{v\}$.
13. **Misprint.** Proof of 6.3, page 60: $y \in Y_2$ becomes $y \in (Y \setminus Y_1) \setminus Y_2$.
14. **Misprint.** Proof of 6.3, page 60: $d(z, x)$ becomes $d(\{w\}, x)$.
15. **Misprint.** Proof of 6.3, page 60: Y_4 becomes $\{v(y) : y \in Y_4\}$, twice.
16. **Misprint.** Proof of 6.3, page 60: $\theta - 1$ becomes θ_1.
17. **Misprint.** Proof of 6.3, page 61: $d(y, v(y)) = \theta_1$ becomes $d(y, v(y)) \geq \theta$.
18. **Misprint.** Proof of 6.3, page 61: $=$ becomes $\geq$, twice.
19. **Misprint.** Proof of 5.1, page 61: $\theta \leq i \leq \delta$ becomes $0 \leq i \leq \delta$.
20. **Misprint.** Proof of 8.3, page 67: $|Y'|$ becomes $|Y_1|$.
21. **Misprint.** Proof of 8.1, page 71: $d(y_1, \sigma)$ becomes $d(y_i, \sigma)$.
22. **Misprint.** Proof of 8.1, page 73: $r \cap U(C) \notin \emptyset$ becomes $r \cap U(C) \neq \emptyset$.
23. **Misprint.** Proof of 8.1, page 74: "which" becomes "with", and $s_2 \notin s_1$ becomes $s_2 \neq s_1$.
24. **Misprint.** Proof of 8.1, page 75: $\geq$ becomes $\leq$ in a triangle inequality, and $d(v, y_1)$ becomes $d(u, y_1)$.
25. **Misprint.** Proof of 8.1, page 75: $d(u, y_j) \geq$ becomes $d(u, y_1) \leq$ in a triangle inequality.

[p. 43] Graph Minors. XVI. Excluding a non-planar graph

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Abstract

This paper contains the cornerstone theorem of the series. We study the structure of graphs with no minor isomorphic to a fixed graph L, when L is non-planar. (The case when L is planar was studied in an earlier paper.) We find that every graph with no minor isomorphic to L may be constructed by piecing together in a tree-structure graphs each of which "almost" embeds in some surface in which L cannot be embedded.

1. Introduction

Graphs in this paper are finite, undirected graphs which may have loops or multiple edges. A graph L is a *minor* of a graph G if L can be obtained from a subgraph of G by contracting edges.

In this paper we are concerned with the structure of the graphs which have no minor isomorphic to a given graph, and we begin in this section with a discussion of what kind of structure we might expect. It turns out that there are (at least) four ingredients to "structure", which we now review.

Ingredient 1: *Tree-structure.* It is a triviality that the graphs with no minor isomorphic to a loop are the forests; this is the simplest instance of the exclusion of a minor forcing some variety of tree-structure. If we exclude K_3 instead we get almost the same answer—forests augmented by loops and multiple edges. Excluding K_4 [p. 44] yields the "series-parallel" graphs, by a theorem of Dirac [1] or Duffin [2], and they too have a certain tree-structure. To unify these and some other instances, we make the following definitions. Let G be a graph with a separation (H_1, H_2) (that is, H_1 and H_2 are subgraphs of G with no common edges and $H_1 \cup H_2 = G$). Let K be a complete graph with $V(K) = V(H_1) \cap V(H_2)$ and with no edges in common with H_1 and H_2. Let $G_i = H_i \cup K$ $(i = 1, 2)$. Then we say that G is the *clique-sum* of G_1 and G_2. If G can be constructed from members of a

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class $\mathcal{C}$ of graphs by (repeated) clique-sums we say that G is a *tree-structure over* $\mathcal{C}$. Thus, the graphs with no K_3-minor are the tree-structures over the class of all graphs with ≤ 2 vertices; and the graphs with no K_4-minor are the tree-structures over the class of all graphs with ≤ 3 vertices. Let us say that G has *tree-width* $\leq N$ if G is a tree-structure over the class of all graphs with $\leq N+1$ vertices (and the smallest such N is of course the *tree-width* of G). The following is the main theorem of [4]. (An *L-minor* is a minor isomorphic to L.)

1.1. *For any planar graph L there is a number N such that every graph with no L-minor has tree-width $\leq N$.*

In a way this generalizes the above-mentioned result about the exclusion of K_3 and K_4. It is not a complete generalization, because 1.1 does not yield a structure which is necessary and sufficient for the exclusion of L, but merely one which is necessary. Nevertheless 1.1 has proved to be very useful. In another sense, 1.1 is best possible, because the structure given by 1.1 is necessary for the exclusion of L, and sufficient for the exclusion of some other (larger) planar graph. (A related fact; if L is non-planar there is no number N as in 1.1.) The object of this paper is to find an analogue of 1.1 for non-planar graphs L, best possible in the same sense; that is, the structure is necessary for the exclusion of L and sufficient for the exclusion of some other, larger graph L'. (In fact it can be shown that for any surface Σ in which L can be drawn, we can choose L' so that it too can be drawn in Σ.)

Ingredient 2: *Genus.* As central to the subject as the genus of L, however, is the genus of graphs with no L-minor. If Σ is a surface and L cannot be drawn in Σ, then no graph which can be drawn in Σ has an L-minor. This provides a second type of structure associated with the exclusion of L as a minor. There are some important theorems about excluded minors which involve combinations of these two structures; for instance, Wagner [12] proved the following. (V_8 is obtained from an eight-vertex circuit by adding edges joining the four opposite pairs of vertices.)

1.2. *A graph G has no K_5 minor if and only if G is a tree-structure over the class of all graphs which are either planar or isomorphic to V_8.*

Ingredient 3: *Bounded extension.* Let us consider the result of excluding K_6. The structure of the graphs with no K_6-minor has not yet been determined, but one class of such graphs consists of those with a vertex the deletion of which yields a planar [p. 45] graph. It can be shown that there is no class of graphs of bounded genus such that all these graphs are tree-structures over it, and that shows the necessity for our third ingredient. Let Σ be a surface, let $N \geq 0$ be an integer, and let L be a graph such that no

graph obtained from L by deleting $\leq N$ vertices can be drawn in Σ. Then there will be no L minor in any graph which can be constructed by adding $\leq N$ vertices (joined arbitrarily) to a graph drawn in Σ. This then yields a third type of structure which we should anticipate. If a subgraph G' of G can be obtained from G by deleting $\leq n$ vertices of G, let us call G a *($\leq n$)-vertex extension* of G'.

We conjectured for some time that these three were sufficient. But that was false.

Ingredient 4: *Vortices.* Take a graph drawn in the plane, and let the vertices on the infinite region be $v_1, v_2, \ldots, v_n$ in order. Add new edges joining v_1 to v_3, v_2 to v_4, v_3 to v_5 and so on, and let the resulting graph be G. Then it can be shown that such a graph has no large clique minor; indeed, Seese and Wessel [11] found the exact bound, that G can have a K_7, but cannot have a K_8, minor. Yet such graphs G cannot be constructed from our first three ingredients, and so a fourth ingredient is needed, so-called "vortices". Roughly, a vortex is a graph with some of its vertices arranged in a circular order, so that however these special vertices are partitioned into two intervals, there are only a bounded number of disjoint paths from one interval to the other. For instance, the graph G above consists of a planar graph together with a vortex (formed by the new edges) inserted into the infinite region.

Our main result implies that these four ingredients suffice. Let us turn to the statement of the theorem. We need first to define what we mean by an "r-ring with perimeter $t_1, \ldots, t_n$". Roughly speaking, it consists a graph G and a sequence $t_1, \ldots, t_n$ of distinct vertices of G, such that G can be constructed as follows. Let $G_1, \ldots, G_n$ be mutually disjoint graphs, each with $\leq r$ vertices, and for $1 \leq i \leq n$ let $t_i \in V(G_i)$. Let us choose some i with $1 \leq i \leq n$, choose $u \in V(G_i)$ and $v \in V(G_{i+1})$ and identify u with v, provided that $t_1, \ldots, t_n$ remain all distinct; and repeat this process as often as we wish.

More precisely, we say G is an *r-ring with perimeter $t_1, \ldots, t_n$* if $t_1, \ldots, t_n \in V(G)$ are distinct and there is a sequence $X_1, \ldots, X_n$ of subsets of $V(G)$, such that

- $X_1 \cup \cdots \cup X_n = V(G)$, and every edge of G has both ends in some X_i,
- $t_i \in X_i$ for $1 \leq i \leq n$,[F1]

[F1] **Misprint.** As printed, all of $t_1, \ldots, t_n$ lie in X_1, which has at most r vertices, so an r-ring could have at most r perimeter vertices. The sets $X_1, \ldots, X_n$ are a path-decomposition laid along the perimeter, one for each t_i.

- $X_i \cap X_k \subseteq X_j$ for $1 \le i \le j \le k \le n$,
- $|X_i| \le r$ for $1 \le i \le n$.

It is easy to check that this is equivalent with the previous definition.

Now let G_0 be a graph drawn in a surface Σ, and let $\Delta_1, \ldots, \Delta_d \subseteq \Sigma$ be pairwise disjoint closed discs, each meeting the drawing only in vertices of G_0, and each containing no vertices of G_0 in its interior. For $1 \le i \le d$ let the vertices of G_0 in $\mathrm{bd}(\Delta_i)$ be $t_1, \ldots, t_n$ say, in order, and choose an r-ring G_i with perimeter $t_1, \ldots, t_n$, meeting G_0 just in $t_1, \ldots, t_n$ and disjoint from every other G_j; and let G be the union of $G_0, G_1, \ldots, G_d$. Such a graph G (and any graph isomorphic to it) is called an *outgrowth by d r-rings of a graph in Σ*. Now we can state our theorem.

[p. 46] **1.3.** *Let L be a nonplanar graph, and let $\Sigma_1, \ldots, \Sigma_s$ be all the connected surfaces (up to homeomorphism) in which L cannot be drawn. Then there are numbers r, d, w such that every graph with no L-minor may be constructed by clique-sums, starting from graphs G' with the following property: G' is a $(\le w)$-vertex extension of an outgrowth by $\le d$ r-rings of a graph that can be drawn in one of $\Sigma_1, \ldots, \Sigma_s$.*

One can think of an r-ring as a graph with a given path-decomposition of bounded width, and in particular, its pieces come in a linear order. Since these pieces are being sewn onto a circular hole in a surface, it would perhaps be more natural if the pieces came in a circular order. One can indeed replace the third condition in the definition of an r-ring by the condition

- $X_i \cap X_k \subseteq X_h \cup X_j$ and $X_h \cap X_j \subseteq X_i \cup X_k$ for $1 \le h \le i \le j \le k \le n$

and 1.3 remains true (because the new statement is obviously weaker than the original); and in fact the new version is equivalent to the original (we leave the equivalence to the reader). This is why we call it a "ring" rather than some more linear name. It is a question of convenience which definition of "r-ring" is used.

While 1.3 has been one of the main goals of this series of papers, it turns out to have been a red herring. There is another result (theorem 3.1) which is proved in this paper, and from which 1.3 is then derived; and in all the future applications in this series of papers, it is not 1.3 but 3.1 that will be needed. Let us explain how theorem 3.1 is used to prove 1.3.

Evidently we would like to eliminate the "tree-structure" part of 1.3 and concentrate on the internal structure of one of the "nodes" of the tree. How can we do so? An inductive argument looks plausible at first sight; if there is no low order cutset of G dividing it into two substantial pieces then G itself must be almost a "node" if the theorem is to be true, while if there is such a cutset we may express G as a clique-sum of two smaller graphs, and hope to apply our inductive hypothesis to these graphs. But there is a

difficulty here; it is possible that these smaller graphs have an L-minor while G does not. Fortunately there is a way to focus in on a "node" which does not involve any decomposing, as follows. We can assume that the tree is as refined as possible in the sense that no node can be split into two smaller nodes, and so for every low order cutset of G, most of any node will lie on one side or the other of the cutset (except for nodes of bounded cardinality, which we can ignore.) Therefore if we fix some node, every small cutset has a "big" side (containing most of the node) and a "small" side—and it turns out that no three small sides have union G. Thus a node defines a "tangle", which is such an assignment of big and small sides to the low order cutsets; and conversely, it can be shown that any tangle in G of sufficiently high "order" will be associated with some node of the tree-structure. Hence a convenient way to analyze the internal structure of the nodes is to analyze the local structure of G with respect to some high order tangle, and this is the content of theorem 3.1.

[p. 47] We have organized the paper backwards, to try to motivate the various steps better. Thus we first prove 1.3 assuming 3.1, then prove 3.1 assuming another statement 4.1, and so on.

2. Structure relative to a tangle

A *separation* of a graph G is a pair (A, B) of subgraphs with $A \cup B = G$ and $E(A \cap B) = \emptyset$, and its *order* is $|V(A \cap B)|$. A *tangle of order θ in G*, where $\theta \geq 1$ is an integer, is a set $\mathscr{T}$ of separations of G, all of order $< \theta$, such that

1. For every separation (A, B) of order $< \theta$, one of (A, B), (B, A) belongs to $\mathscr{T}$
2. If (A_1, B_1), (A_2, B_2), $(A_3, B_3) \in \mathscr{T}$ then $A_1 \cup A_2 \cup A_3 \neq G$, and
3. If $(A, B) \in \mathscr{T}$ then $V(A) \neq V(G)$.

We define $ord(\mathscr{T}) = \theta$.

A *design* is a pair (H, M) where H ~~in~~ is[F2] a graph and M is a set of subsets of $V(H)$. A *location* in G is a set $\{(A_1, B_1), \ldots, (A_k, B_k)\}$ of separations of G such that $A_i \subseteq B_j$ for all distinct i, j with $1 \leq i, j \leq k$. If $\{(A_1, B_1), \ldots, (A_k, B_k)\}$ is a location then

$$(G \cap B_1 \cap \cdots \cap B_k, \{V(A_i \cap B_i) \colon 1 \leq i \leq k\})$$

is a design, which we call the *design of the location*.

[F2] **Misprint.** The word "in" stands for "is".

Let $\theta \geq 1$ be an integer, and let $\mathscr{D}$ be a class of designs. We say that $\mathscr{D}$ is *θ-pervasive in a graph G* if for every subgraph G' of G and every tangle $\mathscr{T}$ in G' of order $\geq \theta$ there is a location $\mathscr{L} \subseteq \mathscr{T}$ with its design in $\mathscr{D}$.

A *tree-decomposition* of a graph G is a pair (T, τ), where T is a tree and for each $t \in V(T)$, $\tau(t)$ is a subgraph of G, such that

- $\cup(\tau(t)\colon t \in V(T)) = G$, and $E(\tau(t) \cap \tau(t')) = \emptyset$ for all distinct $t, t' \in V(T)$,
- if $t, t', t'' \in V(T)$ and t' lies on the path of T between t and t'' then $\tau(t) \cap \tau(t'') \subseteq \tau(t')$.

If (T, τ) is a tree-decomposition of G and $t_0 \in V(T)$, and t_0 has neighbours $t_1, \ldots, t_k \in V(T)$, then

$$(\tau(t_0), \{V(\tau(t_0) \cap \tau(t_i))\colon 1 \leq i \leq k\})$$

is a design, called the *design of t_0 in (T, τ)*.

Let (H, M), (H', M') be designs. We say that (H', M') is an *n-enlargement of (H, M)* if there exists $Z \subseteq V(H')$ such that

- H is a subgraph of H' and $V(H') \setminus V(H) \subseteq Z$,
- every edge of H' is an edge of H,
- for every $X \in M'$ with $X \neq Z$, $X \cap V(H) \in M$,
- n is an integer and $|Z| \leq n$.

If $\mathscr{D}$ is a class of designs and $n \geq 0$ is an integer, the class of all n-enlargements of members of $\mathscr{D}$ is denoted by $\mathscr{D}^n$. The class of all designs (H, M) with $|V(H)| \leq n$ is denoted by $\mathscr{R}_n$. The following is implied by theorem 11.1 of [6].

[p. 48] **2.1.** *For any $\theta \geq 1$, let $\mathscr{D}$ be a class of designs which is θ-pervasive in a graph G. Then G has a tree-decomposition (T, τ) such that for each $t \in V(T)$, the design of t in (T, τ) belongs to $\mathscr{D}^{3\theta-2} \cup \mathscr{R}_{4\theta-3}$.*

Let (H, M) be a design. A graph H' is a *torso* of (H, M) if $V(H) = V(H')$, H is a subgraph of H', and for every $e \in E(H') \setminus E(H)$ there exists $X \in M$ including the ends of e. It is easy to see the following.

2.2. *Let $\mathscr{D}$ be a class of designs, and let $\mathscr{D}'$ be the class of torsos of members of $\mathscr{D}$. If G is a graph with a tree-decomposition (T, τ) such that $\mathscr{D}$ contains the design of t in (T, τ) for each $t \in V(T)$, then G is a tree-structure over $\mathscr{D}'$.*

Proof. For each edge tt' of T, let $K_{tt'}$ be a complete graph with $V(K_{tt'}) = V(\tau(t) \cap \tau(t'))$, chosen so that no two of these complete graphs have a common edge, and none has an edge in common with G. For each subtree S of T, let G_S be the graph with vertex set $\cup(V(\tau(t))\colon t \in V(S))$

and with edge set

$$\cup(E(\tau(t))\colon t \in V(S)) \cup \cup(E(K_{tt'})\colon tt' \in E(T), t \in V(S), t' \notin V(S)).$$

Then $G_T = G$, since $\cup(\tau(t)\colon t \in V(T)) = G$. Hence it suffices to show that G_S is a tree-structure over $\mathscr{D}'$ for every subtree S of T, and we prove this by induction on $|V(S)|$.

Suppose first that $V(S) = \{t_0\}$, and let $t_1, \ldots, t_k$ be the neighbours of t_0 in T. Then $V(G_S) = V(\tau(t_0))$, $\tau(t_0)$ is a subgraph of G_S, and every other edge of G_S is an edge of some $K_{t_0t_i}$, and so its ends belong to $V(\tau(t_0) \cap \tau(t_i))$. Thus G_S is a torso of the design of t_0 in (T, τ), and so $G_S \in \mathscr{D}'$.

Now suppose that $|V(S)| \geq 2$. Let t_0 be a vertex of S with degree one in S, let t_1 be its neighbour in S, and let $S' = S \setminus t_0$. Let H_1 be the graph obtained from $G_{\{t_0\}}$ by deleting the edges of $K_{t_0t_1}$, and let H_2 be obtained from $G_{S'}$ in the same way. We claim that (H_1, H_2) is a separation of G_S with $V(H_1 \cap H_2) = V(K_{t_0t_1})$. Every neighbour t' of t_0 in T with $t' \neq t_1$ is not in $V(S)$, since S is a subtree and t_0 has degree one in S. Hence the edges tt' of T with $t \in V(S)$ and $t' \notin V(S)$ are the edges t_0t' with $t' \neq t_1$, and the edges tt' with $t \in V(S')$, $t' \notin V(S')$ and $t' \neq t_0$. It follows that $H_1 \cup H_2 = G_S$. No edge of $\tau(t_0)$ belongs to $\tau(t)$ with $t \neq t_0$, and the complete graphs $K_{t_0t'}$ ($t' \neq t_1$) appear only in H_1, so H_1 and H_2 have no common edges. Finally, let $t \in V(S')$. The path of T between t_0 and t lies in S, and so it passes through t_1. Hence $\tau(t_0) \cap \tau(t) \subseteq \tau(t_1)$, and so $V(\tau(t_0)) \cap V(\tau(t)) \subseteq V(\tau(t_0) \cap \tau(t_1))$. Since $V(\tau(t_1)) \subseteq V(H_2)$, it follows that $V(H_1 \cap H_2) = V(\tau(t_0) \cap \tau(t_1)) = V(K_{t_0t_1})$. This proves our claim. Since $K_{t_0t_1}$ has no edges in common with H_1 and H_2, and $H_1 \cup K_{t_0t_1} = G_{\{t_0\}}$ and $H_2 \cup K_{t_0t_1} = G_{S'}$, it follows that G_S is the clique-sum of $G_{\{t_0\}}$ and $G_{S'}$. From the inductive hypothesis, $G_{\{t_0\}}$ and $G_{S'}$ are tree-structures over $\mathscr{D}'$, and hence so is G_S. This proves 2.2. $\square$[F3]

Now for $n \geq 0$, if $\mathscr{D}$ is a class of designs then every torso of a member of $\mathscr{D}^n$ is a $(\leq n)$-vertex extension of a torso of a member of $\mathscr{D}$, provided that $Z \cap V(H) = \emptyset$ in the definition of $\mathscr{D}^n$. In general, such a torso G contains a torso G' of a member of $\mathscr{D}$, and $G \setminus X = G' \setminus X$ for some $X \subseteq V(G)$ with $|X| \leq n$.[F4] From 2.1 and 2.2 we deduce the following.

[F3] **Missing proof.** The paper says only that 2.2 is easy to see. The argument above is the one used in the formalization.

[F4] **Statement mistake.** As printed, the sentence fails when Z meets $V(H)$, which page 47 allows, because then a torso of (H', M') can have edges with both ends in $Z \cap V(H)$. For example, let H have two vertices a, b and no edges, and let $M = \emptyset$, $H' = H$, $M' = \{\{a, b\}\}$, $Z = \{a, b\}$ and $n = 2$. The graph with the one edge ab is a torso of (H', M'). The only

2.3. *For any $\theta \geq 1$, let $\mathscr{D}$ be a class of designs which is θ-pervasive in a graph G. Then G may be constructed by clique-sums starting from graphs G' such that either*

- $|V(G')| \leq 4\theta - 3$, *or*
- G' ~~*is a $(\leq 3\theta - 2)$-vertex extension of*~~ *contains a torso G'' of a member of $\mathscr{D}$, and $G' \setminus X = G'' \setminus X$ for some $X \subseteq V(G')$ with $|X| \leq 3\theta - 2$.*[F5]

We shall use 2.3 to derive 1.3 from a theorem that a certain class of designs is θ-pervasive.

3. Surfaces, societies and segregations

In this paper, by a *surface* we mean a non-null compact connected 2-manifold without boundary. An *O-arc* in a surface Σ is a subset $F \subseteq \Sigma$ homeomorphic to a circle. *Open* and *closed discs* in Σ are defined in the natural way. For $X \subseteq \Sigma$, its *closure* is denoted by $\bar{X}$, and $\bar{X} \cap \overline{\Sigma \setminus X}$ is denoted by $bd(X)$. If $F \subseteq \Sigma$ is an O-arc and $X \subseteq F$ is finite then F induces two cyclic permutations on X, called the *natural orders of X from F*.

A *society* is a pair (A, Ω), where A is a graph and Ω is a cyclic permutation of a subset (denoted by $\bar{\Omega}$) of $V(A)$. A *segregation* of G is a set $\mathscr{S}$ of societies such that

- $A \subseteq G$ for every $(A, \Omega) \in \mathscr{S}$, and $\cup(A\colon (A, \Omega) \in \mathscr{S}) = G$,
- $V(A \cap A') \subseteq \bar{\Omega} \cap \bar{\Omega}'$ and $E(A \cap A') = \emptyset$ for all distinct (A, Ω), $(A', \Omega') \in \mathscr{S}$.

We write $V(\mathscr{S}) = \cup(\bar{\Omega}\colon (A, \Omega) \in \mathscr{S})$.

Let Σ be a surface, and $\mathscr{S} = \{(A_1, \Omega_1), \ldots, (A_k, \Omega_k)\}$ a segregation of G. It is convenient always to assume (as we may) that $S \cap V(\mathscr{S}) = \emptyset$. An *arrangement* [p. 49] of $\mathscr{S}$ in Σ is a function α with domain $\mathscr{S} \cup V(\mathscr{S})$, such that (writing $\alpha(A, \Omega)$ for $\alpha((A, \Omega))$):

- for $1 \leq i \leq k$, $\alpha(A_i, \Omega_i)$ is a closed disc $\Delta_i \subseteq \Sigma$, and $\alpha(x) \in bd(\Delta_i)$ for each $x \in \bar{\Omega}_i$,
- for $1 \leq i < j \leq k$, if $x \in \Delta_i \cap \Delta_j$ then $x = \alpha(v)$ for some $v \in \bar{\Omega}_i \cap \bar{\Omega}_j$,
- for all distinct $x, y \in V(\mathscr{S})$, $\alpha(x) \neq \alpha(y)$,

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torso of (H, M) is H, and H is not obtained from that graph by deleting vertices. With the weaker form inserted here, 2.3 holds with the same weakening in its second case, and the proof of 1.3 still goes through, because deleting at most n vertices from an outgrowth by $\leq d$ r-rings gives an outgrowth by $\leq d$ $(n+1)r$-rings. So 1.3 holds as printed, with a larger r than the proof's $2\rho + 1$.

[F5] **Proof gap.** A gap in the proof. The proof of 2.3 uses the sentence before it, which holds only in the weaker form given there, so it proves 2.3 only with the weaker second case given here. We do not know whether 2.3 holds as printed. Its only use is in the proof of 1.3, on page 52, and the weaker form is enough there.

• for $1 \le i \le k$, Ω_i is mapped by α to a natural order of $\alpha(\bar{\Omega}_i)$ from $bd(\Delta_i)$.

An arrangement is *proper* if $\Delta_i \cap \Delta_j = \emptyset$ for all $1 \le i < j \le k$ such that $|\bar{\Omega}_i|, |\bar{\Omega}_j| > 3$. A society (G, Ω) is a *ρ-vortex*, where $\rho \ge 0$ is an integer, if for all distinct $u, v \in \bar{\Omega}$ there do not exist $\rho + 1$ mutually vertex-disjoint paths of G between $I \cup \{u\}$ and $J \cup \{v\}$, where I denotes the set of vertices in $\bar{\Omega}$ after u and before v and J is the set after v and before u, in the natural sense (so $(I \cup \{u\}, J \cup \{v\})$ is a partition of $\bar{\Omega}$). A segregation $\mathscr{S}$ is of *type* (ρ, κ), where $\rho, \kappa \ge 0$ are integers, if $|\bar{\Omega}| > 3$ for at most κ members (A, Ω) of $\mathscr{S}$, and each such member is a ρ-vortex.

Let L be a graph, and let $\mathscr{T}^*$ be a tangle in a graph G. We say that $\mathscr{T}^*$ *controls an L-minor of G* if there is a function α, with domain $V(L) \cup E(L)$, such that

• for each $v \in V(L)$, $\alpha(v)$ is a non-null connected subgraph of G, and for all distinct $u, v \in V(L)$, $\alpha(u)$ and $\alpha(v)$ are vertex-disjoint,

• for each $e \in E(L)$ with distinct ends u, v, $\alpha(e) \in E(G)$ with one end in $V(\alpha(u))$ and the other in $V(\alpha(v))$,

• for every loop $e \in E(L)$ with end v, $\alpha(e) \in E(G) \setminus E(\alpha(v))$ with both ends in $V(\alpha(v))$,

• for all distinct $e, f \in E(L)$, $\alpha(e) \ne \alpha(f)$,

• there is no $(A, B) \in \mathscr{T}^*$ of order $< |V(L)|$ and $v \in V(L)$ such that $V(\alpha(v)) \subseteq V(A)$.

If $Z \subseteq V(G)$, we denote the graph obtained by deleting Z by $G \setminus Z$. If $\mathscr{T}$ is a tangle in G of order θ and $Z \subseteq V(G)$ with $|Z| < \theta$, we denote by $\mathscr{T} \setminus Z$ the set of all separations (A', B') of $G \setminus Z$ of order $< \theta - |Z|$ such that there exists $(A, B) \in \mathscr{T}$ with $Z \subseteq V(A \cap B)$, $A \setminus Z = A'$ and $B \setminus Z = B'$. It is shown in Theorem 8.5 of [6] that $\mathscr{T} \setminus Z$ is a tangle in $G \setminus Z$ of order $\theta - |Z|$.

If $\mathscr{T}$ is a tangle in G, a segregation $\mathscr{S}$ of G is said to be *$\mathscr{T}$-central* if for all $(A, \Omega) \in \mathscr{S}$ there is no $(A', B') \in \mathscr{T}$ with $B' \subseteq A$. Now we can state our main result.

3.1. *For any graph L, there are integers $\kappa, \rho, \zeta \ge 0$ and $\theta \ge 1$ with the following property. Let $\mathscr{T}$ be a tangle of order $\ge \theta$ in a graph G, controlling no L-minor of G. Then there exists $Z \subseteq V(G)$ with $|Z| \le \zeta$, and a $\mathscr{T} \setminus Z$-central segregation of $G \setminus Z$ of type (ρ, κ) which has a proper arrangement in some surface in which L cannot be drawn.*

In the remainder of this section we shall show that 3.1 implies 1.3, by means of 2.3; and then the rest of the paper is devoted to proving 3.1. To deduce 1.3 from 3.1 we need the following lemma.

[p. 50] **3.2.** *Let (G, Ω) be a ρ-vortex, and let the vertices in $\bar{\Omega}$ be $t_1, \ldots, t_n$*

in order, where $n \geq 1$. *Then there are separations* $(A_1, B_1), \ldots, (A_n, B_n)$ *of* G, *such that:*

1. $t_1, \ldots, t_n \in V(B_i)$ *for* $1 \leq i \leq n$,
2. $t_i \in V(A_i)$ *for* $1 \leq i \leq n$,
3. $\mathscr{L} = \{(A_1, B_1), \ldots, (A_n, B_n)\}$ *is a location in* G,
4. (A_i, B_i) *has order* $\leq 2\rho + 1$ *for* $1 \leq i \leq n$, *and*
5. *every torso of the design of* $\mathscr{L}$ *is a* $(2\rho + 1)$*-ring with perimeter* $t_1, \ldots, t_n$.

Proof. By theorem 8.1 of [5], there is a sequence $A_1, \ldots, A_n$ of subgraphs of G, such that:

(a) $A_1 \cup \cdots \cup A_n = G$.
(b) $E(A_i \cap A_j) = \emptyset$ for $1 \leq i < j \leq n$.
(c) $t_i \in V(A_i)$ for $1 \leq i \leq n$.
(d) $A_i \cap A_k \subseteq A_j$ for $1 \leq i \leq j \leq k \leq n$.
(e) $|V(A_i \cap A_j)| \leq \rho$ for $1 \leq i < j \leq n$.

Let A_0 and A_{n+1} be the null graph. For $1 \leq i \leq n$, let B_i be the unique subgraph of G such that (A_i, B_i) is a separation of G and

$$V(A_i \cap B_i) = V(A_{i-1} \cap A_i) \cup V(A_i \cap A_{i+1}) \cup \{t_i\}.$$

(This exists, because it is easy to see that for $i \neq j$ and $1 \leq i, j \leq n$,

$$V(A_i \cap A_j) \subseteq V(A_{i-1} \cap A_i) \cup V(A_i \cap A_{i+1}).)$$

We claim that $(A_1, B_1), \ldots, (A_n, B_n)$ satisfies the theorem.

(1) $A_i \subseteq B_j$ *for* $1 \leq i, j \leq n$ *with* $i \neq j$.

Subproof. To show this it suffices to show that $A_i \cap A_j \subseteq B_j$, since (A_j, B_j) is a separation. But $E(A_i \cap A_j) = \emptyset$ by (b), and

$$V(A_i \cap A_j) \subseteq V(A_{i-1} \cap A_i) \cup V(A_i \cap A_{i+1}) \subseteq V(B_j),$$

and so $A_i \cap A_j \subseteq B_j$. This proves (1).

(2) $t_i \in V(B_j)$ *for* $1 \leq i, j \leq n$.

Subproof. If $i = j$ this is immediate. If $i \neq j$ then again the claim holds, since $t_i \in V(A_i) \subseteq V(B_j)$ by (1). This proves (2).

From (1) and (2) we see that statements 1–4 of 3.2 hold. It remains to show statement 5. For $1 \leq i \leq n$ let $X_i = V(A_i \cap B_i)$. Let H be the subgraph of G with vertex set $X_1 \cup \cdots \cup X_n$ and with no edges; then it is easy to see that $H = B_1 \cap \cdots \cap B_n$. Let $M = \{X_i \colon 1 \leq i \leq n\}$. Then $\mathscr{L}$ has design (H, M).

(3) $X_i \cap X_k \subseteq X_j$ *for* $1 \leq i \leq j \leq k \leq n$.

[p. 51] *Subproof.* Let $v \in X_i \cap X_k$. Then $v \in V(A_i \cap A_k)$, and hence $v \in V(A_j)$ by (d). But we may assume that $j \neq i$, and so $V(A_i) \subseteq V(B_j)$ by (1); and hence $v \in V(B_j)$, and consequently $v \in X_j$. This proves (3).

Since $|X_i| \leq 2\rho + 1$ for $1 \leq i \leq n$, it follows from (3) that every torso of (H, M) is a $(2\rho+1)$-ring with perimeter $t_1, \ldots, t_n$. This proves statement 5 of 3.2, and hence completes the proof of 3.2. □

Proof of 1.3 (assuming 3.1). Let $\theta, \kappa, \rho, \zeta$ be as in 3.1. We may assume (by replacing θ by a larger number if necessary) that $\theta \geq 2\rho + 4 + \zeta$. Let $r = 2\rho + 1$, $d = \kappa$ and $w = \zeta + 4\theta - 3$. We claim that r, d, w satisfy 1.3. For let $\mathcal{D}$ be the class of all designs (G', M) such that every torso of (G', M) is isomorphic to an outgrowth by $\leq d$ r-rings of a graph in one of $\Sigma_1, \ldots, \Sigma_s$, where $\Sigma_1, \ldots, \Sigma_s$ are the surfaces in which L cannot be drawn (up to homeomorphism).

(1) *For any graph G' with no L-minor, $\mathcal{D}^\zeta$ is θ-pervasive in G'.*

Subproof. Let G be a subgraph of G', and let $\mathscr{T}$ be a tangle in G of order $\geq \theta$. Certainly $\mathscr{T}$ controls no L-minor of G, and so, by 3.1, there exists $Z \subseteq V(G)$ with $|Z| \leq \zeta$, and a segregation $\mathcal{S}$ of $G \setminus Z$ of type (ρ, κ), such that

• there is a proper arrangement α of $\mathcal{S}$ in one of $\Sigma_1, \ldots, \Sigma_s$, and • for each $(A, \Omega) \in \mathcal{S}$ there is no $(A', B') \in \mathscr{T}$ with $Z \subseteq V(A' \cap B')$ and $B' \setminus Z \subseteq A$.

Let $(A_1, \Omega_1), \ldots, (A_k, \Omega_k)$ be the members (A, Ω) of $\mathcal{S}$ with $|\bar{\Omega}| > 3$. For $1 \leq j \leq k$, let the vertices of $\bar{\Omega}_j$ be $t^j_1, \ldots, t^j_{n_j}$, in order. Since (A_j, Ω_j) is a ρ-vortex, by 3.2 there are separations $(A^j_1, B^j_1), \ldots, (A^j_{n_j}, B^j_{n_j})$ of A_j as in 3.2. For each $(A, \Omega) \in \mathcal{S}$, let $C(A, \Omega)$ be the unique subgraph B of $G \setminus Z$ such that (A, B) is a separation of $G \setminus Z$ and $V(A \cap B) = \bar{\Omega}$. For $1 \leq j \leq k$ and $1 \leq i \leq n_j$, let $C^j_i = \not{B} B^j_i \cup C(A_j, \Omega_j)$.[F6] Let

$$\mathcal{L} = \{(A^j_i, C^j_i) : 1 \leq j \leq k, 1 \leq i \leq n_j\} \cup \{(A, C(A, \Omega)) : (A, \Omega) \in \mathcal{S}, |\bar{\Omega}| \leq 3\}.$$

Then since $\mathcal{S}$ is a segregation of G, and $(A^j_1, B^j_1), \ldots, (A^j_{n_j}, B^j_{n_j})$ satisfy 3.2, it is straightforward to verify ~~(we omit the details)~~ that $\mathcal{L}$ is a location in $G \setminus Z$ and every torso of the design of $\mathcal{L}$ is an outgrowth by $\leq d$ r-rings of a graph in one of $\Sigma_1, \ldots, \Sigma_s$, that is, $\mathcal{D}$ contains the design of $\mathcal{L}$. (We use here that for $(A, \Omega) \in \mathcal{S}$ with $|\bar{\Omega}| \leq 3$, the edges for the torso of $\mathcal{L}$ with ends in $\bar{\Omega}$ may all be drawn within $\alpha(A, \Omega)$.) We give the details. For $(A, \Omega) \in \mathcal{S}$ with $|\bar{\Omega}| \leq 3$ let $\mathcal{L}(A, \Omega) = \{(A, B_0)\}$, where B_0 is the graph with vertex set $\bar{\Omega}$ and no edges, and for $1 \leq j \leq k$ let $\mathcal{L}(A_j, \Omega_j) = \{(A^j_i, B^j_i) : 1 \leq i \leq n_j\}$. In

[F6] **Misprint.** No B is defined here, and the sets B^j_i are those of the separations (A^j_i, B^j_i) just chosen.

each case $\mathcal{L}(A,\Omega)$ is a location in A (by 3.2 for the vortices), and $\bar{\Omega} \subseteq V(B)$ for each $(A',B) \in \mathcal{L}(A,\Omega)$ (by statement 1 of 3.2). Thus $\mathcal{L}$ is the set of all $(A', B \cup C(A,\Omega))$ with $(A,\Omega) \in \mathcal{S}$ and $(A',B) \in \mathcal{L}(A,\Omega)$. Each of these is a separation of $G \setminus Z$, since (A',B) is a separation of A and $V(A \cap C(A,\Omega)) = \bar{\Omega} \subseteq V(B)$. Since $\mathcal{S}$ is a segregation, $C(A,\Omega)$ includes every other member of $\mathcal{S}$. Hence if two members of $\mathcal{L}$ come from distinct $(A,\Omega),(A^*,\Omega^*) \in \mathcal{S}$, the first side of the first is included in $A \subseteq C(A^*,\Omega^*)$; and if they come from the same (A,Ω), this holds since $\mathcal{L}(A,\Omega)$ is a location. So $\mathcal{L}$ is a location in $G \setminus Z$.

Now let H be a torso of the design (H_0, M) of $\mathcal{L}$. For $(A',B) \in \mathcal{L}(A,\Omega)$ we have $V(A' \cap (B \cup C(A,\Omega))) = V(A' \cap B)$, since $\bar{\Omega} \subseteq V(B)$. Assign each edge of H not in H_0 to some $(A,\Omega) \in \mathcal{S}$ such that its ends lie in $V(A' \cap B)$ for some $(A',B) \in \mathcal{L}(A,\Omega)$, and each edge of H_0 to the member of $\mathcal{S}$ that contains it. For $(A,\Omega) \in \mathcal{S}$, let $P(A,\Omega)$ be the subgraph of H with vertex set $V(H) \cap V(A)$ and with the edges assigned to (A,Ω). Since A is included in every $C(A^*,\Omega^*)$ with $(A^*,\Omega^*) \neq (A,\Omega)$, a vertex of A is in H if and only if it is in $V(B)$ for every $(A',B) \in \mathcal{L}(A,\Omega)$, and the same holds for edges of A and H_0. Hence $P(A,\Omega)$ is a torso of the design of $\mathcal{L}(A,\Omega)$. So $V(P(A,\Omega)) = \bar{\Omega}$ if $|\bar{\Omega}| \leq 3$, and $P(A_j,\Omega_j)$ is a $(2\rho+1)$-ring with perimeter $t^j_1,\ldots,t^j_{n_j}$ by statement 5 of 3.2. Since $\mathcal{S}$ is a segregation, the graphs $P(A,\Omega)$ have union H, no two have a common edge, and $V(P(A,\Omega) \cap P(A^*,\Omega^*)) \subseteq \bar{\Omega} \cap \bar{\Omega}^*$. Also $V(\mathcal{S}) \subseteq V(H)$, since $\bar{\Omega} \subseteq V(B)$ for every $(A',B) \in \mathcal{L}(A,\Omega)$, and so $V(P(A,\Omega)) \cap V(\mathcal{S}) = \bar{\Omega}$.

Let α be the proper arrangement of $\mathcal{S}$ in $\Sigma \in \{\Sigma_1,\ldots,\Sigma_s\}$, and let G_0 be the subgraph of H with vertex set $V(\mathcal{S})$ and with the edges of all $P(A,\Omega)$ with $|\bar{\Omega}| \leq 3$. Then $H = G_0 \cup P(A_1,\Omega_1) \cup \cdots \cup P(A_k,\Omega_k)$. We draw G_0 in Σ with each $x \in V(\mathcal{S})$ at $\alpha(x)$; for $|\bar{\Omega}| \leq 3$ the edges of $P(A,\Omega)$ have ends in $\bar{\Omega}$, and since $|\bar{\Omega}| \leq 3$ they may be drawn in $\alpha(A,\Omega)$ without crossings, meeting $bd(\alpha(A,\Omega))$ only in their ends. This is a drawing, since two of the discs meet only in points $\alpha(v)$ with $v \in V(\mathcal{S})$. Let $\Delta_j = \alpha(A_j,\Omega_j)$ $(1 \leq j \leq k)$. These discs are pairwise disjoint, since α is proper. If $x \in V(\mathcal{S})$ and $\alpha(x) \in \Delta_j$, then $x \in \bar{\Omega}_j$ by the second condition in the definition of an arrangement, and so $\alpha(x) \in bd(\Delta_j)$. Hence Δ_j meets the drawing of G_0 only in the points $\alpha(t^j_i)$, which lie on $bd(\Delta_j)$ in the order $t^j_1,\ldots,t^j_{n_j}$. Finally $P(A_j,\Omega_j)$ meets G_0 just in $\bar{\Omega}_j$, since $V(P(A_j,\Omega_j)) \cap V(\mathcal{S}) = \bar{\Omega}_j$, and it is disjoint from $P(A_{j'},\Omega_{j'})$ for $j' \neq j$, since $\Delta_j \cap \Delta_{j'} = \emptyset$. Thus H is an outgrowth by $k \leq \kappa = d$ r-rings of a graph in Σ.[F7]

[F7] **Missing proof.** The paper omits the verification. The argument above is the one used in the formalization.

For each $(A,B) \in \mathcal{L}$, let (A^+,B^+) be the separation of G such that $Z \subseteq V(A^+ \cap B^+)$, $A^+ \setminus Z = A$, $B^+ \setminus Z = B$ and $E(B^+)$ contains every edge of $E(G) \setminus E(A)$ with both ends in $V(B^+)$. Let $\mathcal{L}^+ = \{(A^+,B^+) : (A,B) \in \mathcal{L}\}$. Then $\mathcal{L}^+$ is a location in G, and its design belongs to $\mathcal{D}^\zeta$, once the separation $(K_e, G \setminus e)$, where K_e consists of e and its ends, is added to $\mathcal{L}^+$ for each edge e of G with an end in Z and all its ends in Z or in the core of $\mathcal{L}$ (such as an edge with both ends in Z, or a loop at a vertex of Z).[F8] To complete the proof of (1) we must show that $\mathcal{L}^+ \subseteq \mathscr{T}$. Let $(A,B) \in \mathcal{L}$, and suppose that $(A^+,B^+) \notin \mathscr{T}$. Now (A,B) has order $\leq \max(3, 2\rho+1)$ from the definition of $\mathcal{L}$, and so (A^+,B^+) has order at most $\max(3,2\rho+1)+\zeta < \theta$. Consequently, $(B^+,A^+) \in \mathscr{T}$. Let $(A',B') = (B^+,A^+)$. Then [p. 52] $Z \subseteq V(A' \cap B')$, and $B' \setminus Z \subseteq A$. But there exists $(A^*,\Omega^*) \in \mathcal{S}$ with $A \subseteq A^*$ from the construction of $\mathcal{L}$, and so $B' \setminus Z \subseteq A^*$. This contradicts the second condition in the definition of $\mathcal{S}$. Hence $\mathcal{L}^+ \subseteq \mathscr{T}$. This completes the proof of (1).

From (1) and 2.3, we deduce that every graph G with no L-minor may be constructed by clique-sums, starting from graphs G' such that either

- $|V(G')| \leq 4\theta - 3 \leq w$, or
- G' ~~is a $(\leq 3\theta-2)$-vertex extension of~~ contains a torso G'' of a member of $\mathcal{D}^\zeta$, and $G' \setminus X = G'' \setminus X$ for some $X \subseteq V(G')$ with $|X| \leq 3\theta - 2$.

But since L is nonplanar, it follows that there is a surface in which L cannot be drawn, and so the null graph is a torso of a member of $\mathcal{D}^\zeta$. Consequently any graph G' satisfying either of the conditions above ~~is a $(\leq w)$-vertex extension of a torso of a member of $\mathcal{D}$~~ contains a torso G'' of a member of $\mathcal{D}$ with $G' \setminus X = G'' \setminus X$ for some $X \subseteq V(G')$ with $|X| \leq w$, and hence is a $(\leq w)$-vertex extension of an outgrowth by $\leq d$ $(w+1)r$-rings of a graph in one of $\Sigma_1, \ldots, \Sigma_s$. This proves 1.3, with $(w+1)r = (w+1)(2\rho+1)$ in place of r.[F9] □

[F8] **Proof gap.** A gap in the proof. Here the core of a location $\{(A_1,B_1),\ldots,(A_k,B_k)\}$ in G is the graph $G \cap B_1 \cap \cdots \cap B_k$ of its design. An n-enlargement (page 47) requires every edge of the larger core to be an edge of the smaller one. Let e be an edge of G with an end in Z and all its ends in Z or in the core of $\mathcal{L}$. Then e is in no A, since $A \subseteq G \setminus Z$, and its ends are in every $V(B^+)$, so e is in every B^+. So e is in the core of $\mathcal{L}^+$ but not in the core of $\mathcal{L}$, which lies in $G \setminus Z$. For example, if G is one edge uz with $Z = \{z\}$ and $\mathcal{L} = \emptyset$, the core of $\mathcal{L}^+$ contains the edge and the core of $\mathcal{L}$ does not. Adding $(K_e, G \setminus e)$ for each such e keeps $\mathcal{L}^+$ a location, because e is in every B^+ and in no A^+, and it removes these edges from the core. The added separations have order at most 2, so they are small by (2.7) of Graph Minors X. A new set $V(K_e)$ meets the core of $\mathcal{L}$ in at most one vertex u, and $u \in V(A \cap B)$ for some $(A,B) \in \mathcal{L}$, because the graphs A with $(A,B) \in \mathcal{L}$ cover $G \setminus Z$ (for a vortex (A_j,Ω_j), by (a) in the proof of 3.2). So once the sets of the design of $\mathcal{L}$ are closed under subsets, with the empty set added, which does not change its torsos, the design of the new location is a $|Z|$-enlargement of it. So 1.3 holds as printed.

[F9] **Proof gap.** A gap in the proof: these two steps used 2.3 and the sentence before it

4. Induction on the surface

The remainder of the paper is devoted to proving 3.1. To get much further in this paper the reader will need to be familiar with [8,9], and so there seems little point in repeating the large number of definitions that we shall need. The reader should therefore see [8] for the meaning of the following terms and notation: drawing, $U(H)$, region, $A(H)$, radial drawing, H-path, respectful tangle, metric of a tangle, free, λ-zone, clearing, rigid, dial, regional distance, battlefield. Also, the reader should see [9] for the terms Σ-span, λ-compression, rearranging within λ of z, (λ, μ)-level. (As these terms turn up in the text, we shall remind the reader again where to look for the definition.)

If $\mathscr{T}^*$ is a tangle in a graph G, we shall usually abbreviate "Σ-span in G with respect to $\mathscr{T}^*$" by "Σ-span" when there is no danger of ambiguity. (For "Σ-span", see [9, section 1]).

If H is a minor of G, and so $E(H) \subseteq E(G)$, and $\mathscr{T}'$ is a tangle in H of order ≥ 2, let $\mathscr{T}$ be the set of all separations (A, B) of G of order $< ord(\mathscr{T}')$ such that there exists $(A', B') \in \mathscr{T}'$ with $E(H) \cap E(A) = E(A')$. By theorem 6.1 of [6], $\mathscr{T}$ is a tangle in G, called the tangle in G induced by $\mathscr{T}'$.

We shall show that 3.1 is implied by the following.

4.1. *Let Σ be a surface and let $p \geq 0$, $\phi \geq 1$. Then there exist $\kappa, \rho, \zeta \geq 0$ and $\theta \geq 1$ such that if $\mathscr{T}^*$ is a tangle in a graph G, and there is a Σ-span of order $\geq \theta$, then either*

- *there is a Σ'-span of order $\geq \phi$, for some surface Σ' obtained from Σ by adding a handle or crosscap, or*
- *there exists $Z \subseteq V(G)$ with $|Z| \leq \zeta$, and a $\mathscr{T}^* \setminus Z$-central segregation of $G \setminus Z$ of type (ρ, κ) with a proper arrangement in Σ, or*
- *$\mathscr{T}^*$ controls a K_p minor of G.*

[p. 53] We remark that in the first outcome of 4.1 there are three possibilities for Σ' (up to homeomorphism) if Σ is orientable, because a handle can be added in two ways, preserving or destroying orientability. If Σ is non-orientable, there are only two possibilities.

To show that 4.1 implies 3.1, we need two lemmas. The $m \times n$ grid has

(page 48) as printed. With their weaker forms, a graph G' in the second case contains a torso G'' of a member of $\mathcal{D}$ with $G' \setminus X = G'' \setminus X$ for some X with $|X| \leq 3\theta - 2 + \zeta \leq w$, and in the first case this holds with G'' null and $X = V(G')$. Deleting at most n vertices from an outgrowth by $\leq d$ r-rings gives an outgrowth by $\leq d$ $(n+1)r$-rings. So $G' \setminus X = G'' \setminus X$ is an outgrowth by $\leq d$ $(w+1)r$-rings, and G' is a $(\leq w)$-vertex extension of it. So 1.3 holds as printed, with the numbers $(w+1)(2\rho+1)$, d and w in place of the r, d, w chosen on page 51.

vertex set

$$\{(i,j) : 1 \le i \le m, 1 \le j \le n\},$$

and (i,j), (i',j') are adjacent if $|i-i'|+|j-j'|=1$. For $1 \le j \le n$, the set of edges joining (i,j) and $(i+1,j)$ for $1 \le i \le m-1$ is called a row of the grid. Our first lemma is as follows.

4.2. *For any integer $\theta \ge 1$ there exists $\theta^* \ge 1$ such that if $\mathscr{T}^*$ is a tangle of order $\ge \theta^*$ in a graph G, there is a Σ-span of order $\ge \theta$, where Σ is the sphere.*

Proof. We may assume that θ is even, by increasing θ if necessary. Let N_0 be the $(2\theta+1) \times \theta$ grid, and let its rows be $P_1, \dots, P_\theta$. Let N_1 be obtained from N_0 by deleting all edges with ends (i,j) and $(i,j+1)$, where $1 \le i \le 2\theta+1$, $1 \le j \le \theta-1$ and $i+j$ is odd. We see that there is a rigid drawing (see [8, sections 2 and 4]) in a sphere, isomorphic to N_1.

Let $\mathscr{T}_1$ be the set of all separations (A,B) of N_1 of order $< \theta$ such that $E(B)$ includes one of $P_1, \dots, P_\theta$.

(1) *$\mathscr{T}_1$ is a tangle in N_1 of order θ.*

Subproof. Let us delete the vertices $(2\theta+1,1), \dots, (2\theta+1,\theta)$ from N_1, and contract the edges joining (i,j) and $(i+1,j)$ with $1 \le i \le 2\theta$ and $1 \le j \le \theta$ such that i is odd. We thereby obtain a graph N_2 isomorphic to the $\theta \times \theta$ grid. Let $\mathscr{T}_2$ be the set of all separations (A',B') of N_2 of order $< \theta$ such that $E(B')$ includes $P_j \cap E(N_2)$ for some j. By theorem 7.3 of [6], $\mathscr{T}_2$ is a tangle of order θ in N_2. But $\mathscr{T}_1$ is the tangle in N_1 induced by $\mathscr{T}_2$. This proves (1).

By theorem 6.1 of [10], there exists $\theta^* \ge 1$ such that

(2) *For every graph G and tangle $\mathscr{T}^*$ in G of order $\ge \theta^*$, there is a $(2\theta+1) \times (2\theta+1)$ grid minor of G such that for all $(A,B) \in \mathscr{T}^*$ of order $\le 2\theta$, $E(A)$ includes no row of the grid.*

We claim that θ^* satisfies the theorem. For let $\mathscr{T}^*$ be a tangle of order $\ge \theta^*$ in a graph G. By (2), G has a $(2\theta+1) \times (2\theta+1)$ grid minor as in (2). Since this grid has a subgraph isomorphic to N_1 (and to simplify notation we may assume that it is N_1) it follows that N_1 is a minor of G, and for every $(A,B) \in \mathscr{T}^*$ of order $\le 2\theta$, $E(A)$ includes none of $P_1, \dots, P_\theta$. Let $\mathscr{T}_3$ be the tangle in G induced by $\mathscr{T}_1$. If $(A,B) \in \mathscr{T}_3$ then $E(B) \cap E(N_1)$ (and hence $E(B)$) includes one of $P_1, \dots, P_\theta$, and so $(B,A) \notin \mathscr{T}^*$. Consequently $(A,B) \in \mathscr{T}^*$, and so $\mathscr{T}_3 \subseteq \mathscr{T}^*$.

[p. 54] Now N_1 has maximum degree ≤ 3, and there is a subgraph N_4 of G isomorphic to a subdivision of N_1 (and hence N_1 is a minor of N_4). We may assume that N_4 is a rigid drawing in a sphere Σ. Let $\mathscr{T}_4$ be the tangle in N_4 induced by $\mathscr{T}_1$. Then $\mathscr{T}_3$ is the tangle in G induced by $\mathscr{T}_4$, and since

$\mathscr{T}_3 \subseteq \mathscr{T}^*$ it follows that $N_4, \mathscr{T}_4$ is a Σ-span of order θ. This proves 4.2. □

Our second lemma is the following, theorem 4.3 of [7].

4.3. *If L is a graph and Σ is a surface such that L can be drawn in Σ, there is an integer $\theta \geq 1$ with the following property. If $\mathscr{T}^*$ is a tangle in a graph G controlling no L minor, then there is no Σ-span of order $\geq \theta$.*

Proof of 3.1 (assuming 4.1). Up to homeomorphism, let $\Sigma_1, \ldots, \Sigma_n$ be all the surfaces on which L cannot be drawn together with the sphere and all the surfaces which can be obtained from a surface on which L cannot be drawn by adding a handle or crosscap. (There are only finitely many up to homeomorphism.) Let us number $\Sigma_1, \ldots, \Sigma_n$ so that for $1 \leq i, j \leq n$, if Σ_j can be obtained from Σ_i by adding a handle or crosscap then $j \geq i$.

Now we define $\kappa_i, \rho_i, \zeta_i$ for $1 \leq i \leq n$ as follows. Choose $p \geq 0$ so that K_p has a minor isomorphic to L. Inductively, suppose that $1 \leq i \leq n$, and $\kappa_j, \rho_j, \zeta_j, \theta_j$ are defined for all j with $i \not\leq < j \leq n$.[F10] If L can be drawn in Σ_i, choose θ_i so that 4.3 is satisfied (with Σ, θ replaced by Σ_i, θ_i) and let $\kappa_i, \rho_i, \zeta_i = 0$. If L cannot be drawn in Σ_i, then $i < n$; choose $\theta_i, \kappa_i, \rho_i, \zeta_i$ so that 4.1 is satisfied (with $\phi, \kappa, \rho, \zeta, \theta$ replaced by $\max(\theta_{i+1}, \ldots, \theta_n), \kappa_i, \rho_i, \zeta_i, \theta_i$). This completes the inductive definition.

Choose $\theta^* \geq 1$ so that 4.2 is satisfied (with θ replaced by θ_1). Let $\kappa = \max(\kappa_1, \ldots, \kappa_n)$, $\rho = \max(\rho_1, \ldots, \rho_n)$, $\zeta = \max(\zeta_1, \ldots, \zeta_n)$ and $\theta = \theta^*$. We claim that $\kappa, \rho, \zeta, \theta$ satisfy 3.1.

For let $\mathscr{T}^*$ be a tangle of order $\geq \theta = \theta^*$ in a graph G, controlling no L-minor. By 4.2, there is a Σ_1-span of order $\geq \theta_1$ (because Σ_1 is the sphere). Choose i with $1 \leq i \leq n$ maximum so that there is a Σ_i-span of order $\geq \theta_i$. If L can be drawn in Σ_i then θ_i satisfies 4.3 and yet $\mathscr{T}^*$ controls no L-minor, a contradiction. Thus, L cannot be drawn in Σ_i, and so one of the outcomes of 4.1 holds (with $\phi, \kappa, \rho, \zeta, \theta$ replaced by $\max(\theta_{i+1}, \ldots, \theta_n), \kappa_i, \rho_i, \zeta_i, \theta_i$). Since L cannot be drawn in Σ_i it follows that every surface which can be obtained from Σ_i by adding a handle or crosscap occurs in the list $\Sigma_{i+1}, \ldots, \Sigma_n$ (up to homeomorphism), and so the first outcome of 4.1 does not hold from the maximality of i. Moreover the third outcome of 4.1 does not hold, since $\mathscr{T}^*$ controls no L-minor and hence controls no K_p-minor, by theorem 4.2 of [7]. Thus the second outcome of 4.1 holds, and so 3.1 holds as required. □

5. Induction on horns

Thus, it remains to prove 4.1. Let Σ be a surface, let $\mathscr{T}^*$ be a tangle in a graph G, and let $H, \eta, \mathscr{T}$ be a Σ-span in G with respect to $\mathscr{T}^*$. Let $v \in V(G) \setminus V(\eta(H))$.

[F10] **Misprint.** Here θ_i is the one being chosen, so only $j > i$ is assumed.

[p. 55] A horn at v of breadth $\geq \theta$ (over $H, \eta, \mathscr{T}$) is a set $\{P_1, \ldots, P_\sigma\}$ of paths of G, such that

- $P_1, \ldots, P_\sigma$ each have one end v and are otherwise mutually disjoint,
- for $1 \leq i \leq \sigma$, P_i has precisely one vertex in $V(\eta(H))$, its end different from v; let this end be $\eta(u_i)$, where $u_i \in V(H)$,
- for $1 \leq i < j \leq \sigma$, $d(u_i, u_j) \geq \theta$ where d is the metric of $\mathscr{T}$ (see [8, section 3]).

Let Σ be a surface. An animal (in G, with respect to $\mathscr{T}^*$) is a quintuple $(H, \eta, \mathscr{T}, X, Y)$, such that $H, \eta, \mathscr{T}$ is a Σ-span, $X \subseteq V(G) \setminus V(\eta(H))$ and $Y \subseteq V(H)$. If $|X| = \chi$ and $|Y| = \delta$, we call this an animal with χ horns and δ hairs. The animal has strength $\geq (\theta, \sigma)$, where $\theta \geq 1$ and $\sigma \geq 0$ are integers, if

- $\mathscr{T}$ has order $\geq \theta$,
- for each $v \in X$ there is a horn at v of cardinality $\geq \sigma$ and breadth $\geq \theta$,
- for each $u \in Y$ there is an $\eta(H)$-path (see [8, section 3]) P with ends $\eta(u), \eta(v)$ say, such that $X \cap V(P) = \emptyset$ and $d(u, v) \geq \theta$, where d is the metric of $\mathscr{T}$,
- $d(u_1, u_2) \geq \theta$ for all distinct $u_1, u_2 \in Y$.

As usual, we shall omit reference to η if η is the identity, and instead speak of the animal $(H, \mathscr{T}, X, Y)$. The animal $(H, \eta, \mathscr{T}, X, Y)$ is hairless if $Y = \emptyset$, and in that case we shall speak of $(H, \eta, \mathscr{T}, X)$ or $(H, \mathscr{T}, X)$. In this section we shall only be concerned with hairless animals.

We shall show that the following implies 4.1.

5.1. *Let Σ be a surface and let $p, \tau, \chi \geq 0$ and $\phi, \psi \geq 1$. Then there exist $\sigma, \kappa, \rho, \zeta \geq 0$ and $\theta \geq 1$ such that if $\mathscr{T}^*$ is a tangle in a graph G, and there is a hairless animal with χ horns, of strength $\geq (\theta, \sigma)$, then either:*

1. *there is a Σ'-span of order $\geq \phi$, for some surface Σ' obtained from Σ by adding a handle or a crosscap, or*
2. *there is a hairless animal with $\chi + 1$ horns, of strength $\geq (\psi, \tau)$, or*
3. *there exists $Z \subseteq V(G)$ with $|Z| \leq \zeta$, and a $\mathscr{T}^* \setminus Z$-central segregation of $G \setminus Z$ of type (ρ, κ) with a proper arrangement in Σ, or*
4. *$\mathscr{T}^*$ controls a K_p-minor of G.*

To prove that 5.1 implies 4.1 we need several lemmas. The first is theorem 4.4 of [7].

5.2. *Let Σ be a surface and let $p \geq 0$. Then there exists $\theta \geq 1$ such that if $\mathscr{T}^*$ is a tangle in a graph G, and $H, \eta, \mathscr{T}$ is a Σ-span of order $\geq \theta$ with metric d, and*

- *$s_1, t_1, \ldots, s_\mu, t_\mu$ are distinct vertices of H, where $\mu = \frac{1}{2}p(p-1)$, such that $d(u, v) \geq \theta$ for all distinct $u, v \in \{s_1, t_1, \ldots, s_\mu, t_\mu\}$, and*

• *$Q_1, \ldots, Q_\mu$ are mutually disjoint $\eta(H)$-paths, such that Q_i has ends $\eta(s_i), \eta(t_i) (1 \le i \le \mu)$,*

then $\mathscr{T}^*$ controls a K_p-minor of G.

[p. 56] We also need

5.3. *Let G be a graph and let $Y \subseteq V(G)$, such that no component of G contains exactly one vertex in Y. If $\mu, \tau \ge 0$ are integers and $|Y| > \tau(\mu - 1)$, then either:*

• *there are μ disjoint paths of G each with both ends distinct and in Y and with no internal vertex in Y, or*

• *there exist $v \in V(G)$ and τ paths $P_1, \ldots, P_\tau$ of G, with ends $v, u_i (1 \le i \le \tau)$ respectively, mutually disjoint except for v, and such that for $1 \le i \le \tau, u_i \in Y$ and $u_i \ne v$, and no internal vertex of P_i is in Y.*

Proof. Let T be a minimal subgraph of G with the properties that $Y \subseteq V(T)$ and any two vertices in Y which belong to the same component of G also belong to the same component of T. We deduce that T is a forest, every component of T contains at least two vertices in Y, and every vertex T with degree 1 in T belongs to Y.

Now let $\mathscr{P}$ be the set of all paths P of T with distinct ends both in Y, and with no internal vertex in Y. Since $\mathscr{P}$ is a set of subtrees of a forest, it follows by an elementary theorem that either there are μ members of $\mathscr{P}$, mutually vertex-disjoint, or there exists $X \subseteq V(T)$ with $|X| < \mu$ meeting all members of $\mathscr{P}$. In the first case we are finished, and so we assume that the second case holds. For each $y \in Y$, X meets the component T' of T containing y, since $|V(T') \cap Y| \ge 2$ and X meets every member of $\mathscr{P}$. Hence there is a minimal path P_y of T from y to some member v_y of X. Since $|Y| > \tau(\mu - 1) \ge \tau|X|$, there exists $x \in X$ and $Y' \subseteq Y$ such that $|Y'| \ge \tau + 1$ and $v_y = x$ for all $y \in Y'$. For each $y \in Y$, no vertex of P_y belongs to X except v_y, and so for distinct $y, y' \in Y'$, P_y and $P_{y'}$ intersect in precisely x (since X meets every path between y and y'). Since there are at least τ members of Y' different from x, it follows that the second outcome of the theorem holds. This proves 5.3. □

5.4. *Let G be a graph, and let $v \in V(G)$. For $1 \le i \le n$ let P_i be a path of G with distinct ends v, u_i, such that $P_1, \ldots, P_n$ are mutually disjoint except for v. Let $Q_1, \ldots, Q_m$ be mutually disjoint paths of G, not passing through v, such that Q_i has ends s_i, t_i for $1 \le i \le m$. Let*

$$Z = \{u_1, \ldots, u_n, s_1, t_1, \ldots, s_m, t_m\}.$$

If $n \ge 5m + 2$, there are $m + 1$ mutually disjoint paths of G, each with distinct ends both in Z.

Proof. Let H be the graph formed by the union of $P_1, \ldots, P_n$ and $Q_1, \ldots, Q_m$; we see that no vertex of H has degree > 4 except possibly v. Let J be the union of all components of $H \setminus v$ which contain at least two vertices in Z, and let $Y = Z \cap V(J)$. Let us apply 5.3 to J, Y, setting $\mu = m + 1$ and $\tau = 5$. Certainly the second statement [p. 57] of 5.3 does not hold, because no vertex of J has degree ≥ 5. If the first statement holds then the theorem is satisfied, and so we assume not. By 5.3, it follows that $|Y| \leq 5m$. Now certainly each Q_j is a path of J because it has at least two vertices in Z, and so $P_i \setminus v \subseteq J$ for each i such that P_i meets some Q_j. Consequently,

$$|Z \setminus Y| = |\{i : 1 \leq i \leq n, P_i \text{ meets no } Q_j\}|.$$

But $|Z \setminus Y| = |Z| - |Y| \geq n - 5m \geq 2$, and so there exist distinct i, i' with $1 \leq i, i' \leq n$ such that $P_i \cup P_{i'}$ meets no Q_j. But then $\{Q_1, \ldots, Q_m, P_i \cup P_{i'}\}$ is a set of $m + 1$ paths satisfying the theorem. This proves 5.4. □

We deduce

5.5. *For any surface Σ and integer $p \geq 0$, there exists $\theta \geq 1$ such that, if $\mathscr{T}^*$ is a tangle in a graph G and there is a hairless animal with $\frac{1}{2}p(p-1)$ horns and with strength $\geq (\theta, 4p(p-1))$, then $\mathscr{T}^*$ controls a K_p-minor of G.*

Proof. Let $\mu = \frac{1}{2}p(p-1)$, and choose $\theta_1 \geq 1$ so that 5.2 is satisfied (with θ replaced by θ_1). Let $\theta = 2\theta_1$: we claim that 5.5 is satisfied. For let $\mathscr{T}^*$ be a tangle in G, and let $(H, \mathscr{T}, X)$ be a hairless animal of strength $\geq (\theta, 8\mu)$ and with $|X| = \mu$.

(1) *For $0 \leq m \leq \mu$ there are m H-paths $Q_1, \ldots, Q_m$, mutually vertex-disjoint, such that*

- *$|V(Q_1 \cup \cdots \cup Q_m) \cap X| \leq m$, and*
- *$d(u, v) \geq \theta_1$ for all distinct $u, v \in \{s_1, t_1, \ldots, s_m, t_m\}$, where Q_i has ends $s_i, t_i (1 \leq i \leq m)$.*

Subproof. We proceed by induction on m. Certainly (1) holds if $m = 0$; we suppose that it holds for some $m < \mu$, and shall prove that it also holds for $m + 1$. Let $Q_1, \ldots, Q_m, s_1, t_1, \ldots, s_m, t_m$ be as in (1).

Since $|(V(Q_1) \cup \cdots \cup V(Q_m)) \cap X| \leq m$, and $|X| = \mu > m$, there exists $x \in X$ with $x \notin V(Q_1), \ldots, V(Q_m)$. Let $\{P_1, \ldots, P_{8\mu}\}$ be a horn at x with breadth $\geq \theta$, and let P_i have ends $x, u_i (1 \leq i \leq 8\mu)$. Since $|X| = \mu$, at most $\mu - 1$ of $P_1, \ldots, P_{8\mu}$ have a vertex in X different from x, and so we may assume that $P_1, \ldots, P_{7\mu}$ have no vertex in $X \setminus \{x\}$. Let $W = \{s_1, t_1, \ldots, s_m, t_m\}$. Now for each $w \in W$ there is at most one i ($1 \leq$

$i \le 7\mu$)[F11] such that $d(u_i, w) < \frac{1}{2}\theta = \theta_1$, since $d(u_i, u_j) \ge \theta$ for all distinct i, j. Consequently, we may assume that for $1 \le i \le 5\mu$, $d(u_i, w) \ge \frac{1}{2}\theta$ for all $w \in W$. Let $n = 5\mu$ and let $G' = P_1 \cup \cdots \cup P_n \cup Q_1 \cup \cdots \cup Q_m$. Now $n = 5\mu \ge 5m + 2$, and so by 5.4 (with G replaced by G') there are $m + 1$ mutually disjoint paths [p. 58] $R_1, \ldots, R_{m+1}$ of G', each with distinct ends both in $\{u_1, \ldots, u_n\} \cup W$, and we may assume that they have no internal vertex in this set. Each of these paths R_i is therefore an H-path since $V(G' \cap H) = \{u_1, \ldots, u_n\} \cup W$. But $d(u, v) \ge \theta_1$ for all distinct $u, v \in \{u_1, \ldots, u_n\} \cup W$. Moreover, $|V(G') \cap X| \le m + 1$, and so

$$|(V(R_1) \cup \cdots \cup V(R_{m+1})) \cap X| \le m + 1.$$

This completes the inductive argument, and hence proves (1).

Consequently, (1) holds with $m = \mu$. Let $Q_1, \ldots, Q_\mu$ be the corresponding paths, and let Q_i have ends $s_i, t_i (1 \le i \le \mu)$. Since 5.2 is satisfied (replacing θ by θ_1) and $d(u, v) \ge \theta_1$ for all distinct $u, v \in \{s_1, t_1, \ldots, s_\mu, t_\mu\}$, it follows from 5.2 that $\mathscr{T}^*$ controls a K_p-minor. This proves 5.5. □

Proof of 4.1 (assuming 5.1). Let Σ be a surface and let $p \ge 0, \phi \ge 1$. Let $\mu = \frac{1}{2}p(p-1)$, and choose θ_μ so that 5.5 holds (with θ replaced by θ_μ). Let $\sigma_\mu = 8\mu$ and $\kappa_\mu = \rho_\mu = \zeta_\mu = 0$. We define $\sigma_i, \theta_i, \kappa_i, \rho_i, \zeta_i$ for $0 \le i \le \mu$ inductively as follows. Suppose that $0 \le i < \mu$ and $\sigma_{i+1}, \theta_{i+1}$ have already been defined. Choose $\sigma_i, \theta_i, \kappa_i, \rho_i, \zeta_i$ so that 5.1 is satisfied (with $\tau, \chi, \psi, \sigma, \kappa, \rho, \zeta, \theta$ replaced by $\sigma_{i+1}, i, \theta_{i+1}, \sigma_i, \kappa_i, \rho_i, \zeta_i, \theta_i$). This completes the inductive definition.

Let $\kappa = \max(\kappa_0, \ldots, \kappa_\mu), \rho = \max(\rho_0, \ldots, \rho_\mu), \zeta = \max(\zeta_0, \ldots, \zeta_\mu)$, and let $\theta = \theta_0$. We claim that $\kappa, \rho, \zeta, \theta$ satisfy 4.1. For let $\mathscr{T}^*$ be a tangle in G with a Σ-span of order $\ge \theta = \theta_0$, and hence with a hairless animal with 0 horns, of strength $\ge (\theta_0, \sigma_0)$. Choose i with $0 \le i \le \mu$ maximum so that there is a hairless animal with i horns, of strength $\ge (\theta_i, \sigma_i)$. If $i = \mu$, then since θ_μ satisfies 5.5, it follows that the third outcome of 4.1 holds. We assume then that $i < \mu$. From the maximality of i, there is no hairless animal with $i + 1$ horns, of strength $\ge (\theta_{i+1}, \sigma_{i+1})$. By 5.1, we deduce that one of outcomes 1, 3, 4 of 5.1 holds, and so 4.1 holds, as required. □

6. Induction on hairs

Next, we show that 5.1 is implied by the following.

6.1. *Let Σ be a surface, and let $p, \tau, \chi, \delta \ge 0$ and $\phi, \psi \ge 1$. Then there exist $\sigma, \kappa, \rho, \zeta \ge 0$ and $\theta \ge 1$ such that if $\mathscr{T}^*$ is a tangle in a graph G, and there is an animal with χ horns and δ hairs, of strength $\ge (\theta, \sigma)$, then either*

[F11] **Misprint.** The letter l for the digit 1.

1. *there is a Σ'-span of order $\geq \phi$, for some surface Σ' obtained from Σ by adding a handle or a crosscap, or*

2. *there is an animal with χ horns and $\delta+1$ hairs, of strength $\geq (\psi, \tau)$, or*

3. *there exists $Z \subseteq V(G)$ with $|Z| \leq \zeta$, and a $\mathscr{T}^* \setminus Z$-central segregation of $G \setminus Z$ of type (ρ, κ) with a proper arrangement in Σ, or*

4. *$\mathscr{T}^*$ controls a K_p-minor of G.*

[p. 59] To show that 6.1 implies 5.1 we need the following lemmas. (For the meaning of "λ-zone", see [8, section 3], and for "λ-compression" see [9, section 1].)

6.2. *Let Σ be a surface and let $\sigma \geq 0$ be an integer. Let $\mathcal{T}^*$ be a tangle in a graph G, and let $H, \mathcal{T}$ be a Σ-span of order θ. Let $\lambda \geq 2$ be an integer with $\theta \geq 4\lambda + 3$, let $\Lambda \subseteq \Sigma$ be a λ-zone such that $H \cap (\Sigma \setminus \Lambda)$ is rigid, and let $\mathcal{T}'$ be the tangle in $H' = H \cap (\Sigma \setminus \Lambda)$ of order $\theta - 4\lambda - 2$ which is a $(4\lambda+2)$-compression of $\mathcal{T}$. Let $v \in V(G) \setminus V(H)$, such that there is a horn at v over $H, \mathcal{T}$ with cardinality $\sigma + 1$ and breadth $\geq \theta$. Then there is a horn at v over $H', \mathcal{T}'$ with cardinality σ and breadth $\geq \theta - 4\lambda - 2$.*

Proof. First, we remark that $\mathcal{T}'$ exists, by theorem 7.10 of [7]. Let $\{P_1, \ldots, P_{\sigma+1}\}$ be a horn at v over $H, \mathcal{T}$, of breadth $\geq \theta$. For $1 \leq i \leq \sigma + 1$ let P_i have ends v, u_i. Now if $u_i, u_j \in \Lambda$ where $i \neq j$, then $d(u_i, u_j) \leq 2\lambda$ since Λ is a λ-zone, and yet $d(u_i, u_j) \geq \theta > 2\lambda$ by hypothesis, a contradiction. Thus, Λ contains at most one of $u_1, \ldots, u_{\sigma+1}$, and hence we may assume that $u_1, \ldots, u_\sigma \notin \Lambda$. Since $\mathcal{T}'$ is the $(4\lambda+2)$-compression of $\mathcal{T}$, it follows that for $1 \leq i < j \leq \sigma$,

$$d'(u_i, u_j) \geq d(u_i, u_j) - 4\lambda - 2 \geq \theta - 4\lambda - 2.$$

Thus $\{P_1, \ldots, P_\sigma\}$ is a horn at v over $H', \mathcal{T}'$ of breadth $\geq \theta - 4\lambda - 2$. This proves 6.2. □

6.3. *Let Σ be a surface, and let $p, \tau, \chi \geq 0$ and $\psi \geq 1$ be integers. Then there exists $\delta \geq 0$ and $\theta \geq 1$ such that if $\mathcal{T}^*$ is a tangle in a graph G, and there is an animal with χ horns and δ hairs of strength $\geq (\theta, \tau + 1)$, then either*

- *there is a hairless animal with $\chi + 1$ horns of strength $\geq (\psi, \tau)$, or*
- *$\mathcal{T}^*$ controls a K_p-minor of G.*

Proof. We may assume that $p \geq 2$, for otherwise the result holds trivially. Let $\mu = \frac{1}{2}p(p-1)$, and choose θ_1 so that 5.2 holds (with θ replaced by θ_1). We may assume, by increasing θ_1, that $\theta_1 \geq \max(\psi, 3)$. Let $\theta = 5\theta_1 + 34$, $\tau' = \max(\tau, 2)$, and $\delta = 2\mu^2\tau'^2$.

We claim that δ, θ satisfy the theorem. For let $\mathcal{T}^*$ be a tangle in a graph G, and let $(H, \mathcal{T}, X, Y)$ be an animal of strength $\geq (\theta, \tau + 1)$, with $|X| = \chi$

and $|Y| = \delta$. For each $y \in Y$ let P_y be an H-path with ends $y, v(y)$, such that $X \cap V(P_y) = \emptyset$ and $d(y, v(y)) \geq \theta$ where d is the metric of $\mathcal{T}$. For each $y \in Y$ let $Q_y = P_y \setminus \{v(y)\}$, and let $G_0 = \bigcup(Q_y : y \in Y)$. Let G_1 be the union of all components of G_0 which contain at least two vertices in Y, and let $Y_1 = Y \cap V(G_1)$. We may assume that $\mathcal{T}^*$ controls no [p. 60] K_p-minor of G. Since $d(y, y') \geq \theta \geq \theta_1$ for all distinct $y, y' \in Y_1$, it follows from 5.2 that:

(1) *There do not exist μ disjoint paths of G_1, each with distinct ends, both in Y_1, and with no internal vertex in Y_1.*

We may assume that:

(2) *There do not exist $v \in V(G_1)$ and τ' paths of G_1 each from v to $Y_1 \setminus \{v\}$, mutually disjoint except for v and with no internal vertex in Y_1.*

Subproof. Otherwise, $v \notin V(H)$ since v has degree $\geq \tau' \geq 2$ in G_1 and every vertex of G_1 in $V(H)$ has degree 1 in G_1. Since $d(y, y') \geq \theta \geq \psi$ for all distinct y, y' it follows that there is a horn at v over $H, \mathcal{T}$ with cardinality τ and breadth $\geq \psi$. But $v \notin X$ since $v \in V(G_1)$ and $X \cap V(G_1) = \emptyset$ (since $X \cap V(P_y) = \emptyset$ for each $y \in Y$) and so $(H, \mathcal{T}, X \cup \{xv\})$[F12] is an animal of strength $\geq (\psi, \tau)$ with $\chi + 1$ horns. Thus we may assume (2).

From (1), (2) and 5.3 applied to G_1, Y_1, we deduce that $|Y_1| \leq (\mu - 1)\tau'$, and so

$$|Y \setminus Y_1| \geq \delta - (\mu - 1)\tau' \geq (2\mu - 1)\mu\tau'.$$

We see that Q_{y_1} is disjoint from Q_{y_2} for all distinct $y_1, y_2 \in Y \setminus Y_1$.

Choose $Y_2 \subseteq Y \setminus Y_1$ maximal such that the set $W = \{y, v(y) : y \in Y_2\}$ has cardinality $2|Y_2|$ and $d(u, v) \geq \theta_1$ for any two distinct $u, v \in W$. By 5.2, we may assume that $|Y_2| \leq \mu - 1$. From the maximality of Y_2 we deduce that for any $y \in (Y \setminus Y_1) \setminus Y_2$[F13] there exists $w \in W$ such that either $d(y, w) < \theta_1$ or $d(v(y), w) < \theta_1$, Consequently there exist $w \in W$ and a set $Y_3 \subseteq Y \setminus Y_1$ with

$$|Y_3| \geq |Y \setminus Y_1|/(2|Y_2|) > \mu\tau'$$

such that for all $y \in Y_3$, either $d(y, w) < \theta_1$ or $d(v(y), w) < \theta_1$. Now $d(y, w) < \theta_1$ for at most one value of $y \in Y$ since $2\theta_1 < \theta$ and $d(y, y') \geq \theta$ for all distinct $y, y' \in Y$. Consequently there exists $Y_4 \subseteq Y_3$ with $|Y_4| \geq \mu\tau'$, such that $d(v(y), w) < \theta_1$ for all $y \in Y_4$.

By theorem 9.2 of [8], there is a $(\theta_1 + 2)$-zone Λ_1 around $\{w\}$ such that $x \subseteq \Lambda_1$ for every $x \in A(H)$ ($A(H)$ is the set of atoms of H—see [8, section 2]) with $d(\{w\}, x) < \theta_1$, since $2 \leq \theta_1 \leq \theta - 3$. Consequently $v(y) \in \bar{\Lambda}_1$

[F12] **Misprint.** Here v is the vertex of the sentence.

[F13] **Misprint.** For $y \in Y_2$ the claim holds with $w = y$, and the maximality of Y_2 is about the other members of $Y \setminus Y_1$.

for all $y \in Y_4$. By theorem 9.3 of [8] there is a $(\theta_1 + 8)$-zone Λ around $\{w\}$ such that $H \cap (\Sigma \setminus \Lambda)$ is rigid and $x \subseteq \Lambda$ for every $x \in A(H)$ with $d(\not{z}\{w\}, x) < \theta_1 + 3$[F14], since $2 \le \theta_1 + 3 \le \theta - 6$. Consequently, $\bar{\Lambda}_1 \subseteq \Lambda$, and so every two vertices in ~~Y_4~~ $\{v(y) : y \in Y_4\}$ are joined by a path P of H with $U(P) \subseteq \Lambda$. (For $U(P)$ see [8, section 2]). Let H_2 be a connected subgraph of H with ~~Y_4~~ $\{v(y) : y \in Y_4\} \subseteq V(H_2)$ and $U(H_2) \subseteq \Lambda$.[F15]

Let $H' = H \cap (\Sigma \setminus \Lambda)$, and let $\mathcal{T}'$ be the $(4(\theta_1+8)+2)$-compression of $\mathcal{T}$ in H' (this exists, by theorem 7.10 of [7]). Then $H', \mathcal{T}'$ is a Σ-span of order $\theta - (4\theta_1 + 34) = ~~\theta - 1~~\theta_1$.[F16]

(3) $Y_4 \subseteq V(H')$.

[p. 61] *Subproof.* Let $y \in Y_4$. Then $d(w, v(y)) < \theta_1$, but $d(y, v(y))~~\not\ge \theta_1~~\theta$ and so $d(w, y) > \theta - \theta_1$.[F17] Since $\theta - \theta_1 \ge \theta_1 + 8$ and Λ is a $(\theta_1 + 8)$-zone, it follows that $y \notin \Lambda$, and so $y \in V(H')$. This proves (3).

Let d' be the metric of $\mathcal{T}'$. Since $\mathcal{T}'$ is a $(4\theta_1+34)$-compression of $\mathcal{T}$, and $d(y_1, y_2)~~\not\ge~~\theta$, it follows that $d'(y_1, y_2)~~\not\ge~~\theta_1$[F18] for all distinct $y_1, y_2 \in Y_4$. Let G_2 be the union of H_2 and all $P_y (y \in Y_4)$. From 5.2, since $H', \mathcal{T}'$ is a Σ-span of order θ_1, it follows that there do not exist μ mutually disjoint paths of G_2 each with distinct ends both in Y_4 and with no internal vertex in Y_4.

Now G_2 is connected and $Y_4 \subseteq V(G_2)$ and $|Y_4| \ge \mu\tau' \ge 2$. From 5.3 (applied to G_2, Y_4), we deduce there exist $v \in V(G_2)$ and τ' paths of G_2 from v to $Y_4 \setminus \{v\}$, mutually disjoint except for v, and each with no internal vertex in Y_4. Since each $y \in Y_4$ has degree 1 in G_2, and $\tau' > 1$, it follows that $v \notin Y_4$ and so $v \in V(G) \setminus V(H')$, since $V(G_2 \cap H') = Y_4$. Consequently there is a horn at v over $H', \mathcal{T}'$, with breadth $\ge \theta_1$ and cardinality τ. But $v \notin X$ since $X \cap V(G_2) = \emptyset$; and from 6.2, for each $x \in X$ there is a horn at x over $H', \mathcal{T}'$ with breadth $\ge \theta_1$ and cardinality τ, since $\theta \ge 4\theta_1 + 35$. Hence $(H', \mathcal{T}', X \cup \{v\})$ is a hairless animal with $\chi + 1$ horns, of strength $\ge (\theta_1, \tau) \ge (\psi, \tau)$, and so the first outcome of 6.3 holds. This proves 6.1. □

[F14] **Misprint.** No z is defined in this proof. The zone Λ is around $\{w\}$, and theorem 9.3 of [8] is applied to the atom $\{w\}$, as theorem 9.2 of [8] is in the sentence before. The next step, $\bar{\Lambda}_1 \subseteq \Lambda$, uses the condition in this form.

[F15] **Misprint.** There are two misprints here. The vertices shown to lie in $\bar{\Lambda}_1 \subseteq \Lambda$ are the far ends $v(y)$ $(y \in Y_4)$, while claim (3) below shows that each $y \in Y_4$ lies outside Λ. With H_2 joining the far ends, $G_2 = H_2 \cup \bigcup_{y \in Y_4} P_y$ is connected and meets H' exactly in Y_4, as the next paragraph needs. So 6.3 holds as printed.

[F16] **Misprint.** Here $\theta = 5\theta_1 + 34$ by its choice on page 59, so $\theta - (4\theta_1 + 34) = \theta_1$. The rest of the proof uses that $H', \mathcal{T}'$ has order θ_1.

[F17] **Misprint.** The paths P_y were chosen on page 59 with $d(y, v(y)) \ge \theta$, and with $d(w, v(y)) < \theta_1$ this gives $d(w, y) > \theta - \theta_1$.

[F18] **Misprint.** Here $\ge$ is meant in both places.

Proof of 5.1 (assuming 6.1). Let Σ be a surface and let $p, \tau, \chi \geq 0$ and $\phi, \psi \geq 1$ be integers. Choose δ and θ_δ so that 6.3 is satisfied (with θ replaced by θ_δ). Let $\kappa_\delta = \rho_\delta = \zeta_\delta = 0$ and $\sigma_\delta = \tau + 1$. For $0 \leq i \leq \delta$ we define $\theta_i, \sigma_i, \kappa_i, \rho_i, \zeta_i$ inductively, as follows. Suppose that $0 \leq i < \delta$ and $\sigma_{i+1}, \theta_{i+1}$ have already been defined. Choose $\theta_i, \sigma_i, \kappa_i, \rho_i, \zeta_i$ so that 6.1 is satisfied (with $\tau, \delta, \psi, \sigma, \kappa, \rho, \zeta, \theta$ replaced by $\sigma_{i+1}, i, \theta_{i+1}, \sigma_i, \kappa_i, \rho_i, \zeta_i, \theta_i$). This completes the inductive definition.

Let $\kappa = \max(\kappa_0, \ldots, \kappa_\delta)$, $\rho = \max(\rho_0, \ldots, \rho_\delta)$, $\zeta = \max(\zeta_0, \ldots, \zeta_\delta)$, $\sigma = \sigma_0$ and $\theta = \theta_0$. We claim that $\sigma, \kappa, \rho, \zeta, \theta$ satisfy 5.1. For let $\mathcal{T}^*$ be a tangle in a graph G, with a hairless animal with χ horns of strength $\geq (\theta, \sigma) = (\theta_0, \sigma_0)$. Choose i with $0 \leq i \leq \delta$ maximum[F19] such that there is an animal with χ horns and i hairs of strength $\geq (\theta_i, \sigma_i)$. If $i = \delta$ then by 6.3, either

- there is a hairless animal with $\chi + 1$ horns, of strength $\geq (\psi, \tau)$, or
- $\mathcal{T}^*$ controls a K_p-minor of G

and in either case 5.1 holds. We assume then that $i < \delta$. From the maximality of i, there is no animal with χ horns and $i+1$ hairs and strength $\geq (\theta_{i+1}, \sigma_{i+1})$. By 6.1, either

- there is a Σ'-span of order $\geq \phi$, for some surface Σ' obtained from Σ by adding a handle or a crosscap, or
- there exist $Z \subseteq V(G)$ with $|Z| \leq \zeta_i \leq \zeta$, and a $\mathcal{T}^* \setminus Z$-central segregation of $G \setminus Z$ of type (ρ_i, κ_i) (and hence of type (ρ, κ)) with a proper arrangement in Σ, or
- $\mathcal{T}^*$ controls a K_p-minor of G.

In each case 5.1 holds, as required. □

[p. 62] 7. The Giant Steps application

The previous paper of this series contains a result designed to be used at this point. We apply it to prove that 6.1 is implied by the following. (For "(λ, μ)-level", see [9, section 8].)

7.1. *Let Σ be a surface, and let $\tau, \chi, \delta, \lambda \geq 0, \theta' \geq 1$ and $\phi, \psi \geq 3$. Then there exist $\theta \geq 1$ and $\sigma \geq 0$ such that if $\mathcal{T}^*$ is a tangle in a graph G, and there is an animal with χ horns and δ hairs, of strength $\geq (\theta, \sigma)$, then either*

- *there is a Σ'-span of order $\geq \phi$, for some surface Σ' obtained from Σ by adding a handle, or*
- *there is an animal with χ horns and $\delta+1$ hairs, of strength $\geq (\psi, \tau)$, or*

[F19] **Misprint.** Here i counts the hairs, and $\theta_i, \sigma_i, \ldots$ were defined for $0 \leq i \leq \delta$. The hairless animal given has $i = 0$.

• *there exists* $Z \subseteq V(G)$ *with* $|Z| \leq \chi + \frac{1}{2}\delta^2\phi^2$ *such that some* Σ*-span of order* $\geq \theta'$ *in* $G \setminus Z$ *with respect to* $\mathcal{T}^* \setminus Z$ *is* $(\lambda, 2\psi)$*-level.*

To encourage the reader, let us point out that we are making progress. 7.1 no longer involves segregations of type (ρ, κ) nor does it involve controlling minors, which were the two fundamental ingredients of 3.1. Also, crosscaps have gone from the first outcome.

To show that 7.1 implies 6.1 we use the following, which is almost theorem 8.4 of [9].

7.2. *Let* Σ *be a surface and let* $p, \phi, \mu \geq 0$*. Then there exist* $\kappa, \lambda, \rho \geq 0$ *and* $\theta \geq 1$ *such that if* $\mathcal{T}^*$ *is a tangle in a graph* G*, and some* Σ*-span of order* $\geq \theta$ *is* (λ, μ)*-level, then either:*

1. *there is a* Σ'*-span of order* ϕ*, for some surface* Σ' *obtained from* Σ *by adding a crosscap, or*

2. *there is a* $\mathcal{T}^*$*-central segregation of* G *of type* (ρ, κ) *with a proper arrangement in* Σ*, or*

3. $\mathcal{T}^*$ *controls a* K_p *minor of* G*.*

However, let us point out a discrepancy. We were not farsighted enough in [9], and omitted to include the term "proper" in the statement of the theorem, although the *proof* in that paper does yield a proper arrangement. (All the arrangements in [9] first come into being via the proof of theorem 7.7 of that paper, so it is enough to check that they are proper at that stage; and they are, as we can see from theorem 7.6 of [9], or from statement (i) of 7.5 in [9].) Alternately, we could pay for the mistake by starting with a non-proper arrangement, and showing how to convert it to a proper one; that is straightforward, but not very short, and it seems unnecessary to inflict it on the reader.

We also need two other lemmas.

[p. 63] **7.3.** *Let* $\mathcal{T}^*$ *be a tangle in a graph* G*, let* $Z \subseteq V(G)$ *with* $|Z| < ord(\mathcal{T}^*)$*, and let* (A, B) *be a separation of* G *of order* $< ord(\mathcal{T}^*) - |Z|$*. Then* $(A, B) \in \mathcal{T}^*$ *if and only if*

$$(A \setminus (Z \cap V(A)), B \setminus (Z \cap V(B))) \in \mathcal{T}^* \setminus Z.$$

Proof. Choose a separation (A^*, B^*) of G such that $Z \subseteq V(A^*) \cap V(B^*)$, and $A^* \setminus Z = A$, $B^* \setminus Z = B$. Then (A^*, B^*) has order at most $|Z|$ more than that of (A, B), and hence less than $ord(\mathcal{T}^*)$. By theorem 2.9 of [6], $(A, B) \in \mathcal{T}^*$ if and only if $(A^*, B^*) \in \mathcal{T}^*$. But $(A^*, B^*) \in \mathcal{T}^*$ if and only if $(A^* \setminus Z, B^* \setminus Z) \in \mathcal{T}^* \setminus Z$. This proves 7.3. □

7.4. *Let* $\mathcal{T}^*$ *be a tangle in* G*, and let* $Z \subseteq V(G)$ *with* $|Z| < ord(\mathcal{T}^*)$*. If* $H, \eta, \mathcal{T}$ *is a* Σ*-span in* $G \setminus Z$ *with respect to* $\mathcal{T}^* \setminus Z$ *then* $H, \eta, \mathcal{T}$ *is a* Σ*-span in* G *with respect to* $\mathcal{T}^*$*.*

Proof. For simplicity we assume that η is the identity. Let $(A, B) \in \mathcal{T}^*$ have order $< ord(\mathcal{T})$; it suffices to show that $(A \cap H, B \cap H) \in \mathcal{T}$. Now

$$|V(A \cap B)| < ord(\mathcal{T}) \leq ord(\mathcal{T}^* \setminus Z) = ord(\mathcal{T}^*) - |Z|$$

and so by 7.3, $(A', B') \in \mathcal{T}^* \setminus Z$, where $A' = A \setminus (Z \cap V(A))$, $B' = B \setminus (Z \cap V(B))$. Since $H, \eta, \mathcal{T}$ is a Σ-span in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$ and (A', B') has order $< ord(\mathcal{T})$, it follows that $(A' \cap H, B' \cap H) \in \mathcal{T}$. But $A \cap H = A' \cap H$ since $Z \cap V(H) = \emptyset$, and similarly $B \cap H = B' \cap H$. Hence $(A \cap H, B \cap H) \in \mathcal{T}$. This proves 7.4. □

Proof of 6.1 (assuming 7.1). Let Σ be a surface, and let $p, \tau, \chi, \delta \geq 0$ and $\phi, \psi \geq 1$. By increasing ϕ, ψ we may assume that $\phi, \psi \geq 3$. Choose $\kappa, \lambda, \rho \geq 0$ and $\theta' \geq 1$ so that 7.2 is satisfied (with θ, μ replaced by $\theta', 2\psi$). Choose θ, σ as in 7.1 and let $\zeta = \lfloor \frac{1}{2}\delta^2\phi^2 \rfloor + \chi$. We claim that 6.1 holds. For let $\mathcal{T}^*$ be a tangle in G, with an animal with χ horns and δ hairs, of strength $\geq (\theta, \sigma)$. Let us apply 7.1. If 7.1.1 or 7.1.2 holds then 6.1.1 or 6.1.2 holds, and so we may assume that 7.1.3 holds, that is,

(1) *There exists $Z \subseteq V(G)$ with $|Z| \leq \zeta$ such that some Σ-span of order $\geq \theta'$ in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$ is $(\lambda, 2\psi)$-level.*

Let us apply 7.2 (with $\theta, \mathcal{T}^*, G, \mu$ replaced by $\theta', \mathcal{T}^* \setminus Z, G \setminus Z, 2\psi$). By (1), the hypotheses of 7.2 are satisfied. We deduce that one of the outcomes of 7.2 holds. But if 7.2.1 holds, that is, there is a Σ'-span of order $\geq \phi$ in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$, then 6.1.1 holds, by 7.4. If 7.2.2 holds then 6.1.3 holds, while if 7.2.3 holds then 6.1.4 holds, because it is easy to see that $\mathcal{T}^*$ controls every minor controlled by $\mathcal{T}^* \setminus Z$. The result follows. □

[p. 64] 8. Non-level Σ-spans

Let $\mathcal{T}$ be a respectful tangle (see [8, section 3]) of order θ in a rigid drawing H in a surface Σ, and let $Y \subseteq V(H)$. If $\gamma \geq 0$, a *γ-envelope around* Y is a family $(\Lambda_y : y \in Y)$ such that:

- for each $y \in Y$, Λ_y is a γ-zone around y,
- for all distinct $y, y' \in Y$, $\bar{\Lambda}_y \cap \bar{\Lambda}_{y'} = \emptyset$,
- for every subset $Y' \subseteq Y$, the drawing $H \cap (\Sigma \setminus \cup(\Lambda_y : y \in Y'))$ is rigid.

(Actually, the third statement here is implied by the same statement for all singleton subsets Y', but it is convenient to present it this way.)

In this section we show that 7.1 is implied by the following. (8.1 will be proved in the next section.)

8.1. *Let Σ be a surface and let $\delta \geq 0$ and $\phi, \psi \geq 3$. Then there exists $\gamma \geq 0$ and $\theta > (4\gamma + 2)\delta$ such that if $\mathcal{T}^*$ is a tangle in a graph G and $H, \mathcal{T}$*

is a Σ-span of order $\geq \theta$, and $Y \subseteq V(H)$ satisfies $|Y| = \delta$ and $d(y, y') \geq \theta$ for all distinct $y, y' \in Y$ (where d is the metric of $\mathcal{T}$), then either:

1. *there is a Σ'-span of order $\geq \phi$ for some surface Σ' obtained from Σ by adding a handle, or*

2. *there is an H-path in G with ends s, t, such that $d(s, t) \geq \psi$ and $d(s, y) \geq \psi$ for all $y \in Y$, or*

3. *there is a γ-envelope $(\Lambda_y : y \in Y)$ around Y, and there exists $Z \subseteq V(G) \setminus V(H')$ with $|Z| \leq \frac{1}{2}\delta^2\phi^2$, such that Z meets every H'-path in G with ends s, t, satisfying $d'(s, t) \geq 2\psi$, where $H' = H \cap (\Sigma \setminus \cup(\Lambda_y : y \in Y))$ and d' is the metric of the $(4\gamma + 2)\delta$-compression of $\mathcal{T}$ in H'.*

To show that 8.1 implies 7.1, we need two lemmas. The first is a kind of converse of 7.4. If $\mathcal{T}$ is a tangle in G and $1 \leq \theta \leq ord(\mathcal{T})$, the set of members of $\mathcal{T}$ of order $< \theta$ is called the *truncation* of $\mathcal{T}$ to order θ.

8.2. *Let $\mathcal{T}^*$ be a tangle in a graph G, and let $H, \mathcal{T}$ be a Σ-span. Let $Z \subseteq V(G) \setminus V(H)$ with $|Z| < ord(\mathcal{T})$, let $1 \leq \theta' \leq ord(\mathcal{T}) - |Z|$, and let $\mathcal{T}'$ be the truncation of $\mathcal{T}$ to order θ'. Then $H, \mathcal{T}'$ is a Σ-span in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$.*

Proof. Since H is a subgraph of $G \setminus Z$ and

$$ord(\mathcal{T}') = \theta' \leq ord(\mathcal{T}) - |Z| \leq ord(\mathcal{T}^*) - |Z| = ord(\mathcal{T}^* \setminus Z),$$

it suffices to show that $(A \cap H, B \cap H) \in \mathcal{T}'$ for all $(A, B) \in \mathcal{T}^* \setminus Z$ of order $< \theta'$. Thus, let $(A, B) \in \mathcal{T}^* \setminus Z$ have order $< \theta'$. By definition of $\mathcal{T}^* \setminus Z$, there exists $(A^*, B^*) \in \mathcal{T}^*$ with $Z \subseteq V(A^* \cap B^*)$, such that $A^* \setminus Z = A$ and $B^* \setminus Z = B$. Then (A^*, B^*) has order

$$|Z| + |V(A \cap B)| < |Z| + \theta' = ord(\mathcal{T})$$

[p. 65] and so $(A^* \cap H, B^* \cap H) \in \mathcal{T}$, since $H, \mathcal{T}$ is a Σ-span with respect to $\mathcal{T}^*$. Now $(A^* \cap H, B^* \cap H) = (A \cap H, B \cap H)$ since $Z \cap V(H) = \emptyset$, and so $(A \cap H, B \cap H) \in \mathcal{T}$; and consequently $(A \cap H, B \cap H) \in \mathcal{T}'$, since it has order $\leq |V(A \cap B)| < ord(\mathcal{T}')$. This proves 8.2. □

Our second lemma is the following.

8.3. *Let Σ be a surface, and let $\tau, \chi, \delta, \lambda, \zeta \geq 0$ and $\psi \geq 3$. Then there exists $\theta > \zeta$ such that if $\mathcal{T}^*$ is a tangle in a graph G, and $(H, \mathcal{T}, X, Y)$ is an animal with χ horns and δ hairs, of strength $\geq (\theta, \zeta + \tau)$, and $Z \subseteq V(G) \setminus V(H)$ with $X \subseteq Z$ and $|Z| \leq \zeta$, then either:*

- *there is an animal with χ horns and $\delta + 1$ hairs of strength $\geq (\psi, \tau)$, or*
- *there is an H-path in $G \setminus Z$ with ends s_1, s_2 and there are distinct $y_1, y_2 \in Y$ with $d(s_1, y_1), d(s_2, y_2) < \psi$, where d is the metric of $\mathcal{T}$, or*

• *$H, \mathcal{T}_0$ is a $(\lambda, 2\psi)$-level Σ-span of order $\theta - |Z|$ in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$, where $\mathcal{T}_0$ is the truncation of $\mathcal{T}$ to order $\theta - |Z|$.*

Proof. Let $\theta = 4\psi + 10\lambda + 3\zeta + 11$. We claim that θ satisfies 8.3. For let $\mathcal{T}^*, G, H, \mathcal{T}, X, Y, Z$ be as in the hypothesis of 8.3, and let d be the metric of $\mathcal{T}$. We may assume that

(1) *$ord(\mathcal{T}) = \theta$.*

Subproof. Let $\mathcal{T}'$ be the truncation of $\mathcal{T}$ to order θ. Then $(H, \mathcal{T}', X, Y)$ is another animal satisfying the same hypotheses, and since $\theta \geq \psi$, it is easy to see that if the theorem is true for $(H, \mathcal{T}', X, Y)$ then it is true for $(H, \mathcal{T}, X, Y)$. Thus it suffices to prove the theorem for $(H, \mathcal{T}', X, Y)$ and (1) follows.

Also, we may assume that

(2) *If there is an H-path in $G \setminus Z$ with ends s_1, s_2 then $d(s_1, s_2) < 2\psi$.*

Subproof. Suppose that $d(s_1, s_2) \geq 2\psi$. If $d(s_1, y) \geq \psi$ for all $y \in Y$, then $(H, \mathcal{T}, X, Y \cup \{s_1\})$ is an animal with χ horns and $\delta + 1$ hairs of strength $\geq (\psi, \tau)$, since $X \subseteq Z$, and so the first outcome of 8.3 holds. We may assume then that $d(s_1, y_1) < \psi$ for some $y_1 \in Y$, and similarly $d(s_2, y_2) < \psi$ for some $y_2 \in Y$. But $y_1 \neq y_2$ since

$$2\psi \leq d(s_1, s_2) \leq d(s_1, y_1) + d(y_1, y_2) + d(y_2, s_2) < 2\psi + d(y_1, y_2)$$

and so the second outcome of 8.3 holds. Thus we may assume (2).

Let $\mathcal{T}_0$ be the truncation of $\mathcal{T}$ to order $\theta - |Z|$. By 8.2, $H, \mathcal{T}_0$ is a Σ-span in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$, and we may assume that it is not $(\lambda, 2\psi)$-level, for otherwise the third outcome of 8.3 holds. But by (2), there is no H-path in $G \setminus Z$ with ends s_1, s_2 such that $d_0(s_1, s_2) \geq 2\psi$, for $d_0(s_1, s_2) \leq d(s_1, s_2)$ (where d_0 is the metric of $\mathcal{T}_0$). Yet [p. 66] $\theta - |Z| \geq 4\lambda + 2\psi + 2$, and so from the definition of $(\lambda, 2\psi)$-level we deduce the following. (For "rearranging" see [9, section 1].)

(3) *There is a Σ-span $H', \eta', \mathcal{T}''$ in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$, of order $\theta - |Z| - 4\lambda - 2$, obtained from $H, \mathcal{T}_0$ by rearranging with λ of some $z \in A(H)$, and there is an $\eta'(H')$-path in $G \setminus Z$ with ends $\eta'(s), \eta'(t)$, such that $d(z, s), d(z, t) \leq \lambda$ and $d''(s, t) \geq 2\psi$, where d'' is the metric of $\mathcal{T}''$.*

Let λ' be an integer with

$$\lambda + \frac{1}{4}|Z| \leq \lambda' < \lambda + \frac{1}{4}|Z| + 1.$$

Now $\mathcal{T}''$ has order $\theta - |Z| - 4\lambda - 2$; let $\mathcal{T}'$ be the truncation of $\mathcal{T}''$ to order $\theta - 4\lambda' - 2$.

(4) *$H', \eta', \mathcal{T}'$ is a Σ-span in G with respect to $\mathcal{T}^*$, obtained from $H, \mathcal{T}$ by rearranging within λ' of z, and $d'(s, t) \geq 2\psi$ where d' is the metric of $\mathcal{T}'$.*

Subproof. Now $H', \eta', \mathcal{T}'$ is a Σ-span in $G \setminus Z$ with respect to $\mathcal{T}^* \setminus Z$, and hence in G with respect to $\mathcal{T}^*$ by 7.4. Let $a \in A(H)$ with $d(a,z) > \lambda'$. Then

$$d_0(a,z) = \min(d(a,z), \theta - |Z|) > \lambda' \geq \lambda$$

and so $z \in A(H')$ (because $H', \eta', \mathcal{T}''$ is obtained from $H, \mathcal{T}_0$ by rearranging within λ of z). Similarly if x is a vertex or edge of H with $d(x,z) > \lambda'$, then $d_0(x,z) > \lambda$ and so $\eta'(x) = x$. Since $\mathcal{T}'$ is the $(4\lambda'+2)$-compression of $\mathcal{T}$ in H', the first claim follows. For the second, we observe that

$$d'(s,t) = \min(d''(s,t), \theta - 4\lambda' - 2) \geq 2\psi.$$

This proves (4).

(5) *For each vertex $w \in V(\eta'(H')) \setminus V(H)$, there is a path Q of $\eta'(H')$ from w to a vertex $x \in V(H)$, such that no vertex of Q belongs to H except x, and $d(z,x) \leq \lambda' + 2$.*

Subproof. Let Q be a minimal path of $\eta'(H')$ from w to $V(H)$, and let its ends be w, x. Then $x \in V(H)$, and no vertex of Q belongs to H except x, by the minimality of Q. Suppose that $d(z,x) \geq \lambda' + 3$. Then $x \in V(H')$ and $\eta'(x) = x$ by (4). Since $x \neq w$, there is an edge e of Q incident with x; let $e = \eta'(f)$ where $f \in E(H')$. Since $\eta'(f)$ is incident with $\eta'(x) = x$, it follows that f is incident with x. Let $a \in A(H)$, with $x \in \bar{a}$ and $a \cap f \neq \emptyset$. Since $x \in \bar{a}$, it follows that $d(a,x) \leq 2$, and so $d(z,a) \geq \lambda' + 1$, since $d(z,x) \geq \lambda' + 3$. By (4), we deduce that $a \in A(H')$ and so $a = f$, since $a \cap f \neq \emptyset$ and $f \in E(H')$. Consequently $f \in E(H)$ and $d(z,f) \geq \lambda' + 1$. By (4) again $e = \eta'(f) = f$, and so $e \in E(H)$, contrary to the minimality of Q. This proves (5).

Let $v \in X$, and let $\{P_1, \ldots, P_{\zeta+\tau}\}$ be a horn at v over $H, \mathcal{T}$ of breadth $\geq \theta$, and let P_i have ends $v, u_i (1 \leq i \leq \zeta + \tau)$.

[p. 67] (6) *$P_1, \ldots, P_{\zeta+\tau}$ may be numbered so that $d(z,u_i) \geq \lambda' + 2 + 2\psi$, and $V(P_i) \cap Z = \{v\}$ for $1 \leq i \leq \tau$.*

Subproof. Since $d(u_i, u_j) \geq \theta$ for $1 \leq i < j \leq \zeta + \tau$, and since $\theta \geq 2(\lambda' + 2 + 2\psi)$, it follows that $d(z,u_i) < \lambda' + 2 + 2\psi$ for at most one value of i. Moreover, since $|Z| \leq \zeta$, there are at most $\zeta - 1$ values of i such that $Z \cap V(P_i) \neq \{v\}$. This proves (6).

(7) *For $1 \leq i \leq \tau$, no internal vertex of P_i belongs to $V(\eta'(H'))$.*

Subproof. Suppose that w is an internal vertex of P_i, where $1 \leq i \leq \tau$ and $w \in V(H')$. Let R be the subpath of P_i between w and u_i, and let Q be a path in $\eta'(H')$ between w and some $x \in V(H)$ with $d(z,x) \leq \lambda' + 2$, such that no vertex of Q is in H except x (this exists by (5).) Then by (6),

$$\lambda' + 2 + 2\psi \leq d(z,u_i) \leq d(z,x) + d(x,u_i) \leq \lambda' + 2 + d(x,u_i)$$

and so $d(x,u_i) \geq 2\psi$. In particular, $x \neq u_i$, and so $Q \cup R$ is an H-path. But $Z \cap V(Q \cup R) = \emptyset$ since $V(Q) \subseteq V(\eta'(H')) \subseteq V(G \setminus Z)$ and $V(R) \subseteq V(P_i) \setminus \{v\} \subseteq V(G) \setminus Z$ by (6). This contradicts (2), and hence proves (7).

(8) *For $1 \leq i \leq \tau, u_i \in V(H')$ and $u_i = \eta'(u_i) \in V(\eta'(H'))$; and for $1 \leq i < j \leq \tau$, $d'(u_i,u_j) \geq \psi$.*

Subproof. By (6), $d(z,u_i) > \lambda'$ and so by (4), $u_i \in V(H')$ and $u_i = \eta'(u_i) \in V(\eta'(H'))$. For the second claim, let $1 \leq i < j \leq \tau$. Then $d(u_i,u_j) \geq \theta$ since $\{P_1,\ldots,P_{\zeta+\tau}\}$ has breadth $\geq \theta$, and so $d'(u_i,u_j) \geq \theta - 4\lambda' - 2$ since $\mathscr{T}'$ is a $(4\lambda'+2)$-compression on $\mathscr{T}$. Since $\theta - 4\lambda' - 2 \geq \psi$, this proves (8).

(9) *For each $v \in X$ there is a horn at v over $H', \eta', \mathscr{T}'$ of cardinality τ and breadth $\geq \psi$.*

Subproof. This follows from (7) and (8), because $v \notin V(\eta'(H'))$ since $v \in X \subseteq Z$ and $V(\eta'(H')) \subseteq V(G \setminus Z)$. This proves (9).

Let $Y_1 = \{y \in Y : d(z,y) \geq \psi + 5\lambda' + 2\}$.

(10) *$|Y_1| \geq \delta - 1$, and $Y_1 \subseteq V(H')$, and $\eta'(y) = y$ for all $y \in Y_1$.*

Subproof. Suppose that $y, y' \in Y \setminus Y_1$ are distinct. Then $d(z,y), d(z,y') \leq \psi + 5\lambda' + 1$, and so $d(y,y') \leq 2(\psi + 5\lambda' + 1) < \theta$, a contradiction. Hence $|Y' Y_1| \geq |Y| - 1 = \delta - 1$.[F20] Now if $y \in Y_1$, then $d(z,y) \geq \psi + 5\lambda' + 2 \geq \lambda'$, and so $y \in V(H')$ and $\eta'(y) = y$ by (4). This proves (10).

Let $Y' = Y \cup \{s,t\}$ (we recall that s and t are defined in (3)). Then $Y' \subseteq V(H')$, by (10).

(11) *$d'(y_1,y_2) \geq \psi$ for all distinct $y_1, y_2 \in Y'$.*

[p. 68] *Subproof.* If $\{y_1,y_2\} = \{s,t\}$ this is true by (3). If $y_1, y_2 \in Y_1$, then

$$d'(y_1,y_2) \geq d(y_1,y_2) - 4\lambda' - 2 \geq \theta - 4\lambda' - 2 \geq \psi$$

since $\mathscr{T}'$ is a $(4\lambda'+2)$-compression of $\mathscr{T}$. Finally, if $y_1 \in Y_1$ and $y_2 \in \{s,t\}$, then

$$\begin{aligned} d'(y_1,y_2) &\geq d'(y_1,z) - d'(y_2,z) \geq d(y_1,z) - 4\lambda' - 2 - d(y_2,z) \\ &\geq (\psi + 5\lambda' + 2) - 4\lambda' - 2 - \lambda \geq \psi \end{aligned}$$

since $d(y_2,z) \leq \lambda$ by (3). This proves (11).

(12) *For all $y \in Y'$ there is an $\eta'(H')$-path in $G \setminus X$ with ends $\eta'(x), \eta'(y)$ say, where $d'(x,y) \geq \psi$.*

Subproof. If $y = s$ or t this is true by (3), since $d'(s,t) \geq 2\psi \geq \psi$ by (4). We assume then that $y \in Y_1$. Since $Y_1 \subseteq Y$ and $(H, \mathscr{T}, X, Y)$ is an animal over $H, \mathscr{T}$ of strength $\geq (\theta, \zeta + \tau)$, it follows that there is an H-path Q in

[F20] **Misprint.** The claim being proved is about Y_1.

$G \setminus X$ with ends x, y say, where $d(x,y) \geq \theta$. Suppose first that some vertex of Q different from y belongs to $\eta'(H')$. Let w be the first such vertex, and let Q' be the subpath of Q between y and w. Let $w = \eta'(y')$ where $y' \in V(H')$. We claim that $d'(y,y') \geq \psi$. For let $a \in A(H)$ with $y' \in a$. If $d(a,z) > \lambda'$ then by (4), $a \in A(H')$ and so $a = \{y'\}$, and $y' \in V(H)$; and by (4) again, $w = \eta'(y') = y'$. Thus $w \in V(H)$, and so $w = x$, since Q is an H-path. But $d(x,y) \geq \theta$, and so $d'(x,y) \geq \theta - 4\lambda' - 2$, since $\mathscr{T}'$ is a $(4\lambda+2)$-compression of $\mathscr{T}$; that is, $d'(y',y) \geq \theta - 4\lambda - 2 \geq \psi$, as claimed.

We may therefore assume that $d(a,z) \leq \lambda'$, and so

$$d(a,y) \geq d(z,y) - d(a,z) \geq (\psi + 5\lambda' + 2) - \lambda' = \psi + 4\lambda' + 2.$$

Since $y' \in a$ and $\mathscr{T}'$ is a $(4\lambda'+2)$-compression of $\mathscr{T}$, it follows that $d(y,y') \geq d(a,y) - 4\lambda' - 2 \geq \psi$, and so again our claim is true. Thus we have shown that $d'(y,y') \geq \psi$, and consequently $y \neq y'$ and y, y' are not adjacent in H' (since $\psi \geq 3$). We deduce that Q' is an $\eta'(H')$-path with ends y, y' (for no vertex of Q' belongs to $\eta'(H')$ except y, w, and $E(Q') \cap E(\eta'(H')) = \emptyset$ even if $|E(Q')| = 1$, since y, y' are not adjacent in H'). Hence (12) holds in this case.

We may therefore assume that y is the only vertex of Q in $\eta'(H')$. Since $x \in V(H)$, there is a minimal path R of H between x and $V(\eta'(H'))$; let its ends be x, w where $w = \eta'(x')$ and $x' \in V(H')$. Suppose that $d'(y,x') < \psi$. Let $a \in A(H)$ with $x' \in a$; then $d(y,a) < \psi + 4\lambda' + 2$ since $\mathscr{T}'$ is a $(4\lambda'+2)$-compression of $\mathscr{T}$. Since $d(z,y) \geq \psi + 5\lambda' + 2$ (because $y \in Y_1$) it follows that $d(z,a) > \lambda'$. By (4), $a \in A(H')$ and so $a = \{x'\}$, and $x' \in V(H)$, and $d(z,x') > \lambda'$; and by (4) again $w = \eta'(x') = x'$. Let e be the edge of R incident with w. Then $e \in E(H)$, and $d(z,e) \geq d(z,x') > \lambda'$. By (4), $e \in E(H')$ and $\eta'(e) = e$, and so both ends of e belong to $V(\eta'(H'))$, contrary to the minimality of R. We deduce that $d'(y,x') \geq \psi$. Hence $y \neq x'$ and so $Q \cup R$ is an $\eta'(H')$-path, satisfying (12). This proves (12).

From (9)–(12) we see that $(H', \eta', \mathscr{T}', X, Y')$ is an animal with χ horns and $\geq \delta + 1$ hairs, and so the first outcome of 8.3 holds. This proves 8.3. □

[p. 69] **Proof of 7.1** (assuming 8.1). Let Σ be a surface, and let $\tau, \chi, \delta, \lambda \geq 0, \theta' \geq 1$, and $\phi, \psi \geq 3$. Choose $\gamma \geq 0$ and $\theta_1 > (4\gamma+2)\delta$ so that 8.1 holds (with θ replaced by θ_1). Let $\zeta = \lfloor \frac{1}{2}\delta^2\phi^2 \rfloor + \chi$, and choose $\theta_2 > \zeta$ so that 8.3 holds (with θ replaced by θ_2). We may assume that $\theta_2 \geq \max(\theta' + \zeta, 4\psi + 2)$. Let $\sigma = \delta + \tau + \zeta$ and $\theta = \chi + \theta_1 + 3\gamma + (4\gamma+2)\delta + \theta_2$.

We claim that σ, θ satisfy 7.1. For let $\mathscr{T}^*$ be a tangle in G, and let $(H, \mathscr{T}, X, Y)$ be an animal with χ horns and δ hairs, of strength $\geq (\theta, \sigma)$. Let $\mathscr{T}'$ be the truncation of $\mathscr{T}$ to order $\theta - \chi$. By 8.2, $H, \mathscr{T}'$ is a Σ-span

in $G \setminus X$ with respect to $\mathscr{T}^* \setminus X$. Let d' be the metric of $\mathscr{T}'$. From 7.4 we deduce

(1) *If there is a Σ'-span of order $\geq \phi$ in $G \setminus X$, for some surface Σ' obtained from Σ by adding a handle, then 7.1.1 holds.*

Next, we observe

(2) *If there is an H-path in $G \setminus X$ with ends s, t such that $d'(s,t) \geq \psi$ and $d'(s,y) \geq \psi$ for all $y \in Y$, then 7.1.2 holds.*

Subproof. Then $d(s,t) \geq d'(s,t)$, and $d(s,y) \geq d'(s,y)$ for all $y \in Y$, and so $(H, \mathscr{T}, X, Y \cup \{s\})$ is an animal (in G, with respect to $\mathscr{T}^*$) with χ horns and $\delta + 1$ hairs, of strength $\geq (\psi, \sigma) \geq (\psi, \tau)$. This proves (2).

Now $\mathscr{T}'$ has order $\theta - \chi \geq \theta_1$, and for all distinct $y_1, y_2 \in Y$,

$$d'(y_1, y_2) = \min(d(y_1, y_2), \theta - \chi) \geq \min(\theta, \theta - \chi) = \theta - \chi \geq \theta_1$$

and so 8.1 may be applied (with $\mathscr{T}^*, G, \mathscr{T}, \theta$ replaced by $\mathscr{T}^* \setminus X, G \setminus X, \mathscr{T}', \theta_1$). From (1) and (2) we may therefore assume that 8.1.3 holds, that is,

(3) *There is a γ-envelope $(\Lambda_y : y \in Y)$ around Y, and there exists $Z \subseteq V(G \setminus X) \setminus V(H'')$ with $|Z| \leq \frac{1}{2}\delta^2\phi^2$, such that Z meets every H-path in $G \setminus X$ with ends s, t satisfying $d''(s,t) \geq 2\psi$, where $H'' = H \cap (\Sigma \setminus \bigcup(\Lambda_y : y \in Y))$, and d'' is the metric of the $(4\gamma + 2)\delta$-compression $\mathscr{T}''$ of $\mathscr{T}'$ in H''.*

Now $(H, \mathscr{T}', X, Y)$ is an animal of strength $\geq (\theta - \chi, \sigma)$, and $H'', \mathscr{T}''$ is obtained from $H, \mathscr{T}'$ by repeating δ times the operation of clearing a γ-zone (see [8, section 3]). From δ applications of 6.2 we deduce

(4) *For each $v \in X$ there is a horn at v over $H'', \mathscr{T}''$ of cardinality $\sigma - \delta = \tau + \zeta$ and breadth $\geq \theta - \chi - (4\gamma + 2)\delta \geq \theta_2$.*

For each $y \in Y$, let P_y be a minimal path of H between y and $V(H \cap bd(\Lambda_y))$, with ends $y, p(y)$ say.

(5) *$d''(p(y_1), p(y_2)) \geq \theta_2$ for all distinct $y_1, y_2 \in Y$.*

[p. 70] *Subproof.* Now

$$\begin{aligned} \theta - \chi &\leq d'(y_1, y_2) \leq d'(y_1, p(y_1)) + d'(p(y_1), p(y_2)) + d'(p(y_2), y_2) \\ &\leq 2\gamma + d'(p(y_1), p(y_2)) \end{aligned}$$

since for $i = 1, 2$, Λ_{y_i} is a γ-zone around y_i and $p(y_i) \in \bar{\Lambda}_{y_i}$. But

$$d'(p(y_1), p(y_2)) \leq d''(p(y_1), p(y_2)) + (4\gamma + 2)\delta$$

since $\mathscr{T}''$ is a $(4\gamma + 2)\delta$-compression of $\mathscr{T}'$. Consequently,

$$d''(p(y_1), p(y_2)) \geq \theta - \chi - 2\gamma - (4\gamma + 2)\delta \geq \theta_2.$$

This proves (5).

(6) *For each $y \in Y$ there is a H''-path in $G \setminus X$ with ends $p(y), q(y)$ say, such that $d''(p(y), q(y)) \geq \theta_2$.*

Subproof. Let Q_y be an H-path in $G \setminus X$ with ends $y, r(y)$ say, such that $d'(y, r(y)) \geq \theta - \chi$. Let R_y be a minimal path of H between $r(y)$ and $V(H')$, with ends $r(y), q(y)$ say. Then for some $y' \in Y, \bar{\Lambda}_{y'}$ contains both $r(y)$ and $q(y)$, and so

$$d'(y', r(y)), d'(y', q(y)) \leq \gamma.$$

Hence $d'(q(y), r(y)) \leq 2\gamma$. But $d'(y, p(y)) \leq \gamma$ since $p(y) \in \bar{\Lambda}_y$, and so

$$\begin{aligned} \theta - \chi \leq d'(y, r(y)) &\leq d'(y, p(y)) + d'(p(y), q(y)) + d'(q(y), r(y)) \\ &\leq 3\gamma + d'(p(y), q(y)). \end{aligned}$$

Thus $d'(p(y), q(y)) \geq \theta - \chi - 3\gamma \geq \theta_2$, and so $P_y \cup Q_y \cup R_y$ is an H''-path satisfying (6). This proves (6).

Let $Y'' = \{p(y) \colon y \in Y\}$. From (4)–(6) it follows that $(H'', \mathscr{T}'', X, Y'')$ is an animal in G with respect to $\mathscr{T}^*$ with χ horns and δ hairs, of strength $\geq (\theta_2, \zeta + \tau)$. By 8.3 (with $\theta, H, \mathscr{T}, Y, Z$ replaced by $\theta_2, H'', \mathscr{T}'', Y'', Z \cup X$) we deduce that one of the outcomes of 8.3 holds.

If the first outcome of 8.3 holds, then 7.1.3 is true. Suppose that the second outcome of 8.3 holds, that is, there is an H''-path P in $G \setminus (X \cup Z)$ with ends s_1, s_2 and there are distinct $y_1, y_2 \in Y''$ with $d''(s_1, y_1), d''(s_2, y_2) < \psi$. Then P is an H-path in $G \setminus X$, not meeting Z, and so $d''(s_1, s_2) < 2\psi$ by (3). Hence

$$d''(y_1, y_2) \leq d''(y_1, s_1) + d''(s_1, s_2) + d'(s_2, y_2) < 4\psi \leq \theta_2$$

contrary to (5). Thus this case does not occur. We may therefore assume that the third outcome of 8.3 holds, that is, there is a $(\lambda, 2\psi)$-level Σ-span $H'', \mathscr{T}_0$ of order $\theta_2 - |X \cup Z|$ in $G \setminus (X \cup Z)$ with respect to $\mathscr{T}^* \setminus (X \cup Z)$. But then 7.1.3 holds, since $\theta_2 - |X \cup Z| \geq \theta_2 - \zeta \geq \theta'$. In all cases, therefore, 7.1 holds, as required. □

[p. 71] 9. The main proof

Now we complete the series of reductions by proving 8.1.

Proof of 8.1. Let Σ be a surface, and let $\delta \geq 0$ and $\phi, \psi \geq 3$ be integers. Let $\kappa = \lfloor \frac{1}{2}\phi^2\delta^2 \rfloor + 1$, let β be even with $\kappa(\psi + 6) \leq \beta \leq \kappa(\psi + 6) + 1$, let $\gamma = 3\beta + 3$, and let $\theta = 12(\delta + 3)(\beta + 1)$. We claim that γ, θ satisfy 8.1. For let $\mathscr{T}^*$ be a tangle in a graph G, let $H, \mathscr{T}$ be a Σ-span of order $\geq \theta$

with metric d, and let $Y \subseteq V(H)$ with $|Y| = \delta$, such that $d(y, y') \geq \theta$ for all distinct $y, y' \in Y$. We may assume that 8.1.2 is false, that is,

(1) *There is no H-path with ends s, t such that $d(s,t) \geq \psi$ and $d(s,y) \geq \psi$ for all $y \in Y$.*

Let $Y = \{y_1, \ldots, y_\delta\}$. For $1 \leq i \leq \delta$ there is a battlefield (Λ_i, C_i, X_i) around y_i of size β (see [8, section 9]), by theorem 9.5 of [8], since $\beta \geq 4$ is even and $\theta \geq 16\beta + 17$. We may assume that $|X_i| = \beta$ for each i.

For $1 \leq i < j \leq \delta$, if $\sigma \in \bar{\Lambda}_i \cap \bar{\Lambda}_j$ then $d(y_i, \sigma), d(y_j, \sigma) \leq 3\beta + 3$,[F21] and so

$$\theta \leq d(y_i, y_j) \leq d(y_i, \sigma) + d(y_j, \sigma) \leq 6\beta + 6,$$

a contradiction. Thus the discs $\bar{\Lambda}_1, \ldots, \bar{\Lambda}_\delta$ are mutually disjoint.

For $i = 0, 1, \ldots, \delta$, let $H_i = H \cap (\Sigma \setminus (\Lambda_1 \cup \cdots \cup \Lambda_i))$.

(2) *For $0 \leq i \leq \delta$, H_i is a rigid drawing in Σ.*

Subproof. Certainly $H_0 = H$ is rigid, and so we may assume that $i \geq 1$. Let $F \subseteq \Sigma$ be an O-arc with $F \cap U(H_i) \subseteq V(H_i)$ and $|F \cap V(H_i)| \leq 2$. Since the circuits $H \cap bd(\Lambda_1), \ldots, H \cap bd(\Lambda_i)$ are mutually disjoint and $|F \cap V(H_i)| \leq 2$, and $i \geq 1$, it follows that for some $j (1 \leq j \leq i)$, the sets $F \cap \Lambda_1, \ldots, F \cap \Lambda_i$ are all empty except possibly $F \cap \Lambda_j$. Consequently, if H' denotes $H \cap (\Sigma \setminus \Lambda_j)$, then $F \cap U(H') \subseteq V(H')$ and $|F \cap V(H')| \leq 2$. But H' is rigid, since (Λ_j, C_j, X_j) is a battlefield, and so there is a dial Δ for F, H' (see [8, section 4]). Now Δ includes none of $bd(\Lambda_1), \ldots, bd(\Lambda_i)$ since it includes no circuit of H', and F meets none of $\Lambda_1, \ldots, \Lambda_i$ except possibly Λ_j, and it follows that Δ is disjoint from all of $\Lambda_1, \ldots, \Lambda_i$ except possibly Λ_j. Consequently, Δ is a dial for F, H_i. This proves (2).

Let $\mathscr{T}_0 = \mathscr{T}$, and for $i = 1, \ldots, \delta$ let $\mathscr{T}_i$ be the tangle obtained from $\mathscr{T}_{i-1}$ by clearing the $(3\beta + 3)$-zone Λ_i. (We observe that Λ_i is indeed a $(3\beta + 3)$-zone with respect to $\mathscr{T}_{i-1}$ since it is for $\mathscr{T}_0$, and thus this clearing operation is possible since $\mathscr{T}_{i-1}$ has order $\geq \theta - (12\beta + 14)(i-1) > 12\beta + 14$. We see that $\mathscr{T}_i$ has order $ord(\mathscr{T}) - (12\beta + 14)i$ and is the $(12\beta + 14)i$-compression of $\mathscr{T}$ in H_i.

For "free", see [8, section 3].

(3) *For $1 \leq i \leq \delta$, X_1 is free with respect to $\mathscr{T}_i$.*

[p. 72] *Subproof.* This is true for $i = 1$, since (Λ_1, C_1, X_1) is a battlefield. We proceed by induction on i, and suppose that $1 < i \leq \delta$, and that X_1 is free with respect to $\mathscr{T}_{i-1}$. Let d_{i-1} be the metric of $\mathscr{T}_{i-1}$. Then for each

[F21] **Misprint.** The next display uses $d(y_i, \sigma)$.

$v \in X_1$,

$$\begin{aligned} d_{i-1}(y_i, v) &\geq d_{i-1}(y_i, \Lambda_1) - 1 \geq d(y_i, y_1) - (12\beta + 14)(i-1) - 1 \\ &\geq \theta - (12\beta + 14)(\delta - 1) - 1 > 2\beta + 5(3\beta + 3) + 2 \end{aligned}$$

and so X_1 is free with respect to $\mathscr{T}_i$, by theorem 4.7 of [9] (with $\mathscr{T}, H, z, \lambda, X, H'$ replaced by $\mathscr{T}_{i-1}, H_{i-1}, y_i, 3\beta + 3, X_1, H_i$). This proves (3).

By theorem 4.1 of [9] there is at most one $(12\beta + 14)i$-compression of $\mathscr{T}$ in H_i, and so $\mathscr{T}_i$ is unique. In particular, $\mathscr{T}_i$ does not depend on the order in which $\Lambda_1, \ldots, \Lambda_i$ are cleared. From this observation and (2), (3), we deduce that for $1 \leq i \leq j \leq \delta$, X_i is free with respect to $\mathscr{T}_j$, and $(\Lambda_i : 1 \leq i \leq \delta)$ is a γ-envelope around Y. Let $H' = H_\delta, \mathscr{T}' = \mathscr{T}_\delta$.

(4) *For $1 \leq i < j \leq \delta$, if there are ϕ^2 mutually disjoint H'-paths in G with one end in X_i and the other in X_j, then 8.1.1 holds.*

Subproof. Let $P_1, \ldots, P_{\phi^2}$ be such a set of paths, and let P_k have ends $a_k \in X_i, b_k \in X_j$ where $a_1, \ldots, a_{\phi^2}$ are in order in $bd(\Lambda_i)$. By a theorem of Erdös and Szekeres [3] there exist $1 \leq i_1 < i_2 < \cdots < i_\phi \leq \phi^2$ such that $b_{i_1}, b_{i_2}, \ldots, b_{i_\phi}$ are in order in $bd(\Lambda_2)$ (under one of the orientations of $bd(\Lambda_2)$). The claim follows from theorem 3.5 of [9], since $\phi \geq 3$. This proves (4).

In view of (4), we may assume that there exists Z_0 such that

(5) *$Z_0 \subseteq V(G), Z_0 \cap V(H') \subseteq X_1 \cup \cdots \cup X_\delta, |Z_0| < \kappa$, and Z_0 meets every H'-path with ends in distinct X_i, X_j.*

Subproof. Let $1 \leq i < j \leq \delta$. By (4) and Menger's theorem, we may assume that there exists $Z_{ij} \subseteq V(G)$ with $Z_{ij} \cap V(H') \subseteq X_i \cup X_j$ and $|Z_{ij}| \leq \frac{1}{2}\phi^2$, such that Z_{ij} meets every H'-path with ends in X_i and X_j. Set $Z_0 = \cup(Z_{ij} : 1 \leq i < j \leq \delta)$; then Z_0 satisfies (5).

We suppose, for a contradiction, that setting $Z = Z_0 \cap (V(G) \setminus V(H'))$ does not satisfy 8.1.3. Consequently, if d' is the metric of $\mathscr{T}'$, it follows that there is an H'-path P_0 in G with ends s_1, s_2, such that:

(6) *$Z_0 \cap V(P_0) \subseteq \{s_1, s_2\} \cap (X_1 \cup \cdots \cup X_\delta)$ and $d'(s_1, s_2) \geq 2\psi$.*

For $i = 1, 2$ let t_i be the second vertex of P_0 in $V(H)$ as P_0 is traversed from s_i (s_i is the first).

(7) *$d(s_1 t_1) < \psi$ and $d(s_2, t_2) < \psi$.*

[p. 73] *Subproof.* If s_1, t_1 are adjacent in P_0, joined by an edge of H, then $d(s_1, t_1) \leq 2 < \psi$, so we assume not. Then the subpath of P_0 between s_1 and t_1 is an H-path. But for $1 \leq i \leq \delta$, since $s_1 \notin \Lambda_i$ it follows from the definition of a battlefield that $d(s_1, y_i) \geq \beta \geq \psi$. Consequently, $d(s_1, t_1) < \psi$ by (1). This proves (7).

We may assume that

(8) $t_1 \in \Lambda_1$ *and* $t_2 \in \Lambda_2$.

Subproof. Now $t_1 \neq s_2$ since $d(s_1, s_2) \geq d'(s_1, s_2) \geq 2\psi$ and $d(s_1, t_1) < \psi$. Consequently, $t_1 \in \Lambda_1 \cup \cdots \cup \Lambda_\delta$ since P_0 is an H'-path. Similarly, $t_2 \in \Lambda_1 \cup \cdots \cup \Lambda_\delta$. If for some i, $\{t_1, t_2\} \subseteq \Lambda_i$, then by (7), $d'(s_1, \Lambda_i) \leq d(s_1, t_1) < \psi$ and similarly $d'(s_2, \Lambda_i) < \psi$, and so $d'(s_1, s_2) < 2\psi$ contrary to (6). Thus t_1, t_2 belong to distinct members of $\Lambda_1, \ldots, \Lambda_\delta$, and we may assume from the symmetry that $t_1 \in \Lambda_1$ and $t_2 \in \Lambda_2$. This proves (8).

We may assume that

(9) *There is no path* Q *of* H *from* X_1 *to* $V(P_0)$ *with* $V(Q) \cap V(H') \subseteq X_1$ *and with* $Z_0 \cap V(Q) = \emptyset$.

Subproof. Suppose that for $i = 1, 2$, Q_i is a path of H from X_i to $V(P_0)$ with $V(Q_i) \cap V(H') \subseteq X_i$ and with $Z_0 \cap V(Q_i) = \emptyset$. The union of Q_1, Q_2 and P_0 includes a path P of G from X_1 to X_2 with $V(P) \cap V(H') \subseteq X_1 \cup X_2$ and with $Z_0 \cap V(P) = \emptyset$. But this contradicts (5). Thus either Q_1 or Q_2 does not exist, and without loss of generality we may assume the former. This proves (9).

For $3 \leq i \leq \beta - 3$ there is an $(i+2)$-zone M_i around y_1 including every $x \in A(H)$ with $d(y_1, x) < i$, since $\theta > \beta$.

(10) *There are* κ *mutually disjoint paths of* H *between* $V(H) \cap bd(M_\kappa)$ *and* $V(H) \cap bd(\Lambda_1)$.

Subproof. Suppose not. By a form of Menger's theorem for planar graphs, there is a circuit C of a radial drawing (see [8, section 2]) K of H, of length $< 2\kappa$ with $U(C) \subseteq \bar{\Lambda}_1$, bounding an open disc in $\bar{\Lambda}_1$ including M_κ. By theorem 7.5 of [8], $ins(C) \subseteq \bar{\Lambda}_1$ since

$$|E(C)| < 2\kappa < 2(\theta - (3\beta + 3)).$$

Let r be a region of H with $r \cap U(C) \neq \emptyset$ (see [8, section 2]). Then $d(r, y_1) \leq \frac{1}{2}|E(C)| < \kappa$ since $r \cap U(C) \not\in \neq \emptyset$[F22] and $y_1 \in ins(C)$; and so $r \subseteq M_\kappa$ by the choice of M_κ. But then r is a subset of the open disc bounded by $U(C)$, and so $r \cap U(C) = \emptyset$, a contradiction. This proves (10).

[p. 74] (11) *There is a path* P_1 *of* H *between* $V(H) \cap bd(M_\kappa)$ *and* X_1, *with only one vertex in* $V(H')$ *and such that* $Z_0 \cap V(P_1) = \emptyset$.

Subproof. We recall that (Λ_1, C_1, X_1) is a battlefield of size β, and hence $\bar{M}_\kappa$ is included in the open disc in Λ_1 bounded by $U(C_1)$. Moreover, the regional distance between $U(C_1)$ and $bd(\Lambda_1)$ (see [8, section 5]) is at least $1 + \frac{1}{2}\beta$, and so from (10) and theorem 10.5 of [5], there are κ disjoint paths of H between $V(H) \cap bd(M_\kappa)$ and X_1, each with only one vertex in $V(H')$. Since $|Z_0| < \kappa$, one of these paths is disjoint from Z_0. This proves (11).

[F22] **Misprint.** The sentence before shows the intended text.

(12) *For $3 \le i < j \le \beta - 3$ if $j - i \ge 3$ then $\bar{M}_i \subseteq M_j$.*

Subproof. Now $d(y_1, x) \le i + 2$ for every $x \in A(H)$ with $x \subseteq \bar{M}_i$, and since $i + 2 < j$ it follows that $x \subseteq M_j$. This proves (12).

Let i be such that $3 \le i \le \beta - \psi - 5$. Then M_i and $M_{i+\psi+2}$ are both defined, and $\bar{M}_i \subseteq M_{i+\psi+2}$ by (12). Consequently $\Sigma_i = \bar{M}_{i+\psi+2} \setminus M_i$ is homeomorphic to a closed cylinder. Let J_i be the subdrawing of H formed by the vertices and edges of H in Σ_i. Now the cylinders $\Sigma_{\kappa+r(\psi+5)}$ $(r = 0, \dots, \kappa - 1)$ all exist since $\kappa + (\kappa - 1)(\psi + 5) \le \beta - \psi - 5$, and by (12) they are mutually disjoint. Consequently the graphs $J_{\kappa+r(\psi+5)}$ $(r = 0, \dots, \kappa - 1)$ are mutually disjoint, and since $|Z_0| < \kappa$, there exists i with $\kappa \le i \le \beta - \psi - 5$ such that $Z_0 \cap V(J_i) = \emptyset$.

(13) *$\bar{M}_\kappa \subseteq M_{i+\psi+2}$, and $M_{i+\psi+2} \subseteq \Lambda_1$.*

Subproof. Now $i \ge \kappa$ and $\psi \ge 1$, so the first inclusion follows from (12). For the second, let $x \in A(H)$ ~~which~~ with[F23] $x \subseteq \bar{M}_{i+\psi+2}$. Then $d(y_1, x) \le i + \psi + 4$ by definition of $M_{i+\psi+2}$; and so $x \subseteq \Lambda_1$ since Λ_1, C_1, X_1 is a battlefield of size β around y_1 and $i + \psi + 4 < \beta$. This proves (13).

We recall that P_0 was defined before (6).

(14) *$P_0 \cap J_i$ is null.*

Subproof. Now H is connected, and both O-arcs in $bd(\Sigma_i)$ correspond to circuits of J_i, and so J_i is connected. Let P_1 be as in (11); then P_1 has one end in $bd(M_\kappa)$ and the other in $bd(\Lambda_1)$, and so from (13), P_1 meets $H \cap bd(M_{i+\psi+2}) \subseteq J_i$. Suppose that P_0 also meets J_i. Now $U(J_i) \subseteq \Lambda_1$ by (13), and so there is a path of $P_1 \cup J_i$ from X_1 to $V(P_0)$ with only one vertex in H'. But $V(P_1 \cup J_i) \cap Z_0 = \emptyset$, contrary to (9). This proves (14).

Let v be the first vertex of P_0 (as P_0 is traversed from s_1) such that $v \in V(H), v \ne s_1$ and $v \notin \Lambda_1 \setminus M_i$. (Such a vertex exists, because $s_2 \in V(H), s_2 \not\in \not\neq s_1$ and $s_2 \notin \Lambda_1 \setminus M_i$.) Let u [p. 75] be the last vertex of P_0 in $V(H)$ before v. (This exists, because $s_1 \in V(H)$ is before v.) Let P be the subpath of P_1 between u and v.

(15) *$i + \psi + 2 \le d(u, y_1) \le 3\beta + 3 + \psi$.*

Subproof. Now $u \notin M_i$ and hence $u \notin M_{i+\psi+2}$ by (14). The first inequality follows from the choice of $M_{i+\psi+2}$. For the second, it follows from (7) and (8) if $u = s_1$, because

$$d(s_1, y_1) \not\ge \le d(s_1, t_1) + d(t_1, y_1) < \psi + 3\beta + 3.$$[F24]

If $u \ne s_1$, then $u \in \Lambda_1$ by the choice of v and so $d(\not v u, y_1) \le 3\beta + 3$. This proves (15).

[F23] **Misprint.** In the same way, $s_2 \notin s_1$ below becomes $s_2 \ne s_1$.

[F24] **Misprint.** A misprint: the triangle inequality gives $\le$. In the next sentence, $d(u, y_1)$ is meant, since u is the vertex in Λ_1.

From (15), $d(u, y_1) \geq \psi$; and for $2 \leq j \leq \delta$,

$$d(u, y_j) \geq d(y_1, y_j) - d(u, y_1) \geq \theta - (3\beta + 3 + \psi) \geq \psi$$

by (15). Since P is an H-path with ends u, v, it follows from (1) that $d(u, v) < \psi$. Now if $v \in M_i$ then $d(y_1, v) \leq i + 2$, and so

$$d(u, y_j) \not\geq \leq d(u, v) + d(y_1, v) < \psi + i + 2,$$ [F25]

contrary to (15). Thus $v \notin M_i$, and so $v \notin \Lambda_1$ from the definition of v. But $v \neq s_2$ by (1), and so $v \in \Lambda_j$ for some j with $2 \leq j \leq \delta$, since P_0 is an H'-path. Consequently $d(v, y_j) \leq 3\beta + 3$. Hence

$$\theta \leq d(y_1, y_j) \leq d(y_1, u) + d(u, v) + d(v, y_j) < (3\beta + 3 + \psi) + \psi + (3\beta + 3) < \theta,$$

a contradiction. Thus, our assumption that 8.1.3 is not satisfied was false. This proves 8.3. □

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[F25] **Misprint.** A misprint: the triangle inequality bounds $d(u, y_1)$ from above, and this contradicts (15).

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